

BDA30703

Signal Flow Graphs

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Outline

- Introduction to Signal Flow Graphs
 - Definitions
 - Terminologies
 - Examples
- Mason's Gain Formula
 - Examples
- Signal Flow Graph from Block Diagrams
- Design Examples



Introduction

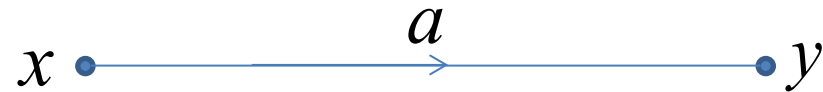
- Alternative method to block diagram representation, developed by Samuel Jefferson Mason.
- Advantage: the availability of a flow graph gain formula, also called Mason's gain formula.
- A signal-flow graph consists of a network in which nodes are connected by directed branches.
- It depicts the flow of signals from one point of a system to another and gives the relationships among the signals.

Fundamentals of Signal Flow Graphs

- Consider a simple equation below and draw its signal flow graph:

$$y = ax$$

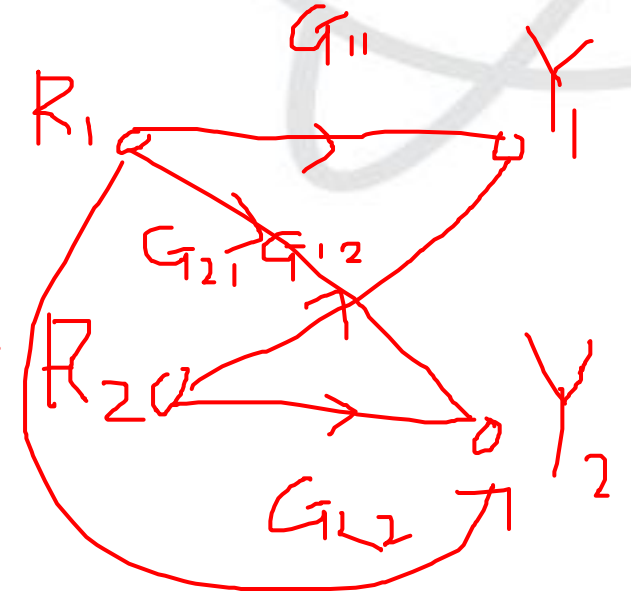
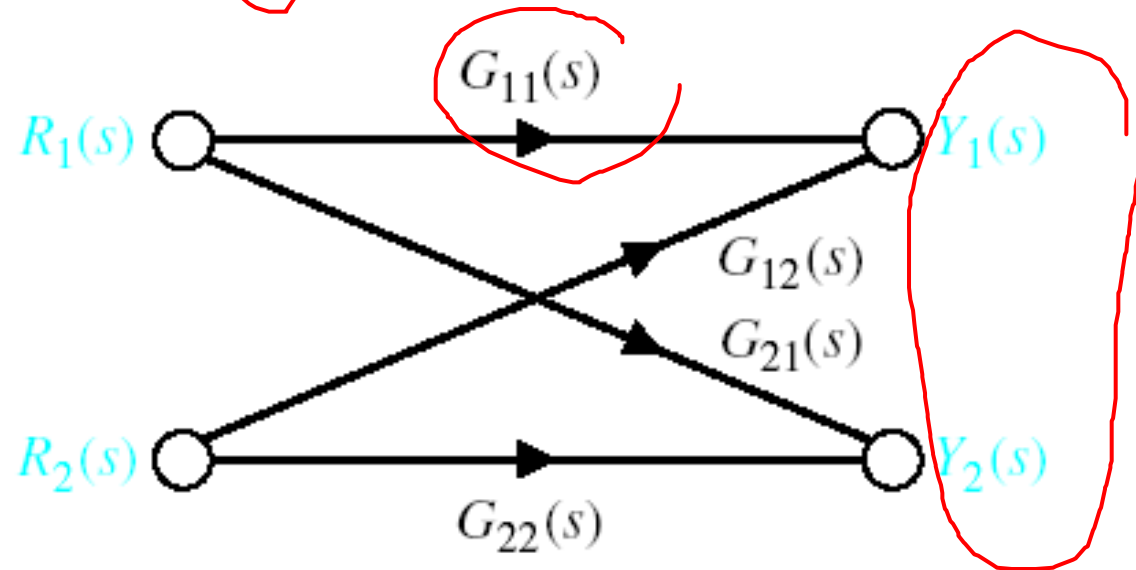
- The signal flow graph of the equation is shown below;



- Every variable in a signal flow graph is designed by a **Node**.
- Every transmission function in a signal flow graph is designed by a **Branch**.
- Branches are always **unidirectional**.
- The arrow in the branch denotes the **direction** of the signal flow.

Signal-Flow Graph Models

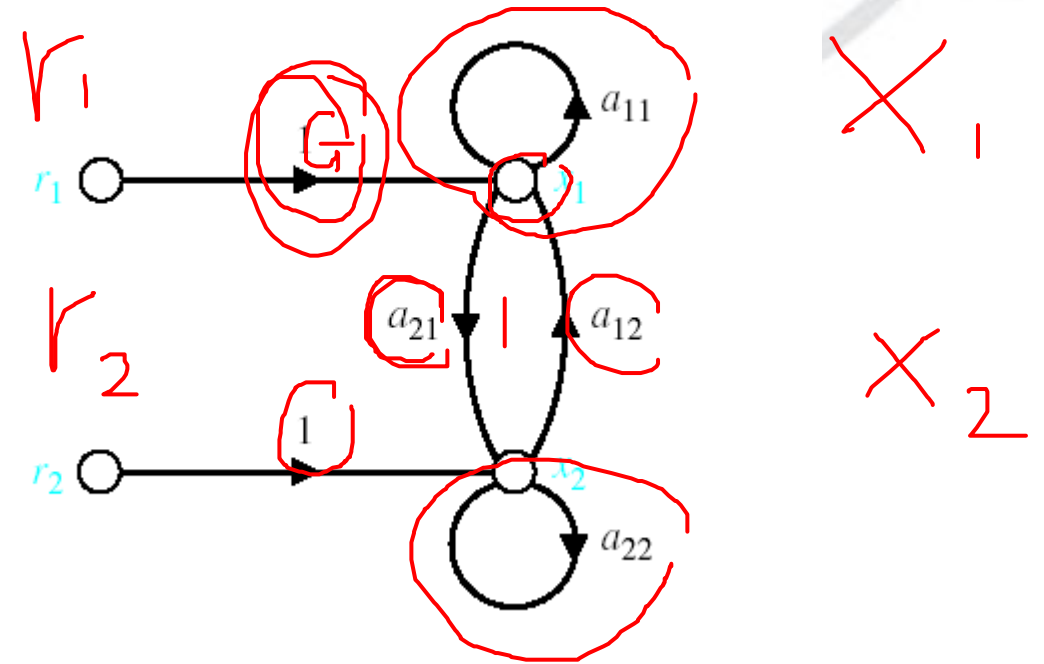
$$\begin{aligned} \rightarrow & \quad Y_1(s) = G_{11}(s) \cdot R_1(s) + G_{12}(s) \cdot R_2(s) \\ \rightarrow & \quad Y_2(s) = G_{21}(s) \cdot R_1(s) + G_{22}(s) \cdot R_2(s) \end{aligned}$$



Signal-Flow Graph Models

r_1 and r_2 are inputs and x_1 and x_2 are outputs

$$\begin{aligned} \rightarrow & \quad a_{11} \cdot x_1 + a_{12} \cdot x_2 + \overset{G}{\parallel} r_1 = x_1 \\ \rightarrow & \quad \cancel{a_{21}} \cdot x_1 + a_{22} \cdot x_2 + \parallel r_2 = x_2 \end{aligned}$$



Signal-Flow Graph Models

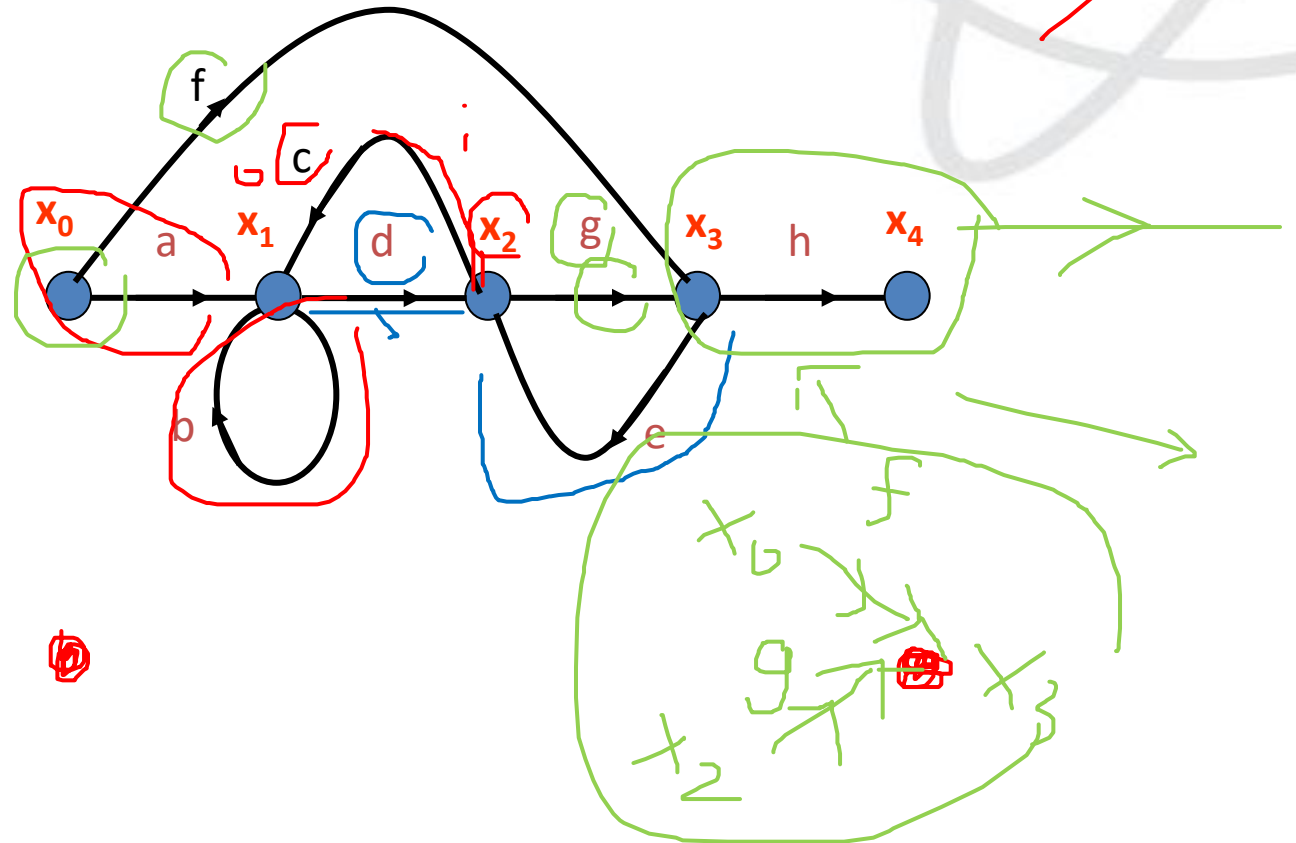
- x_0 is input and x_4 is output

→ $x_1 = ax_0 + bx_1 + cx_2$

→ $x_2 = dx_1 + ex_3$

→ $x_3 = fx_0 + gx_2$

→ $x_4 = hx_3$



x_1 x_2 x_3 x_4

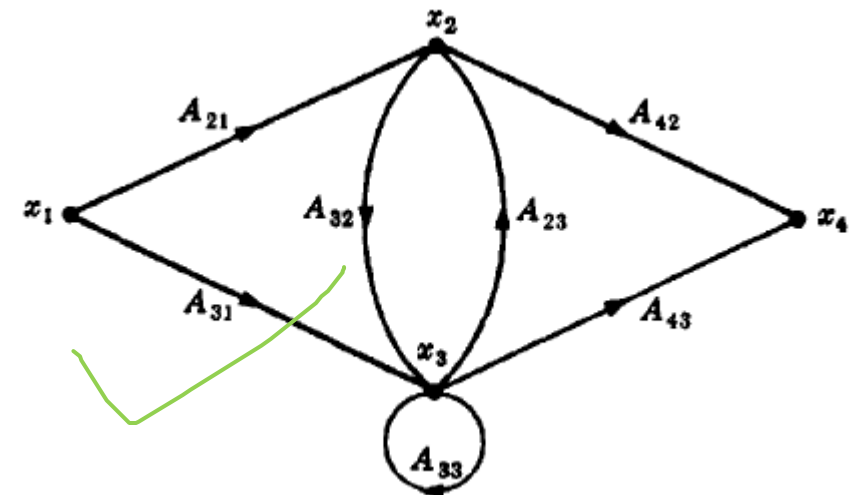
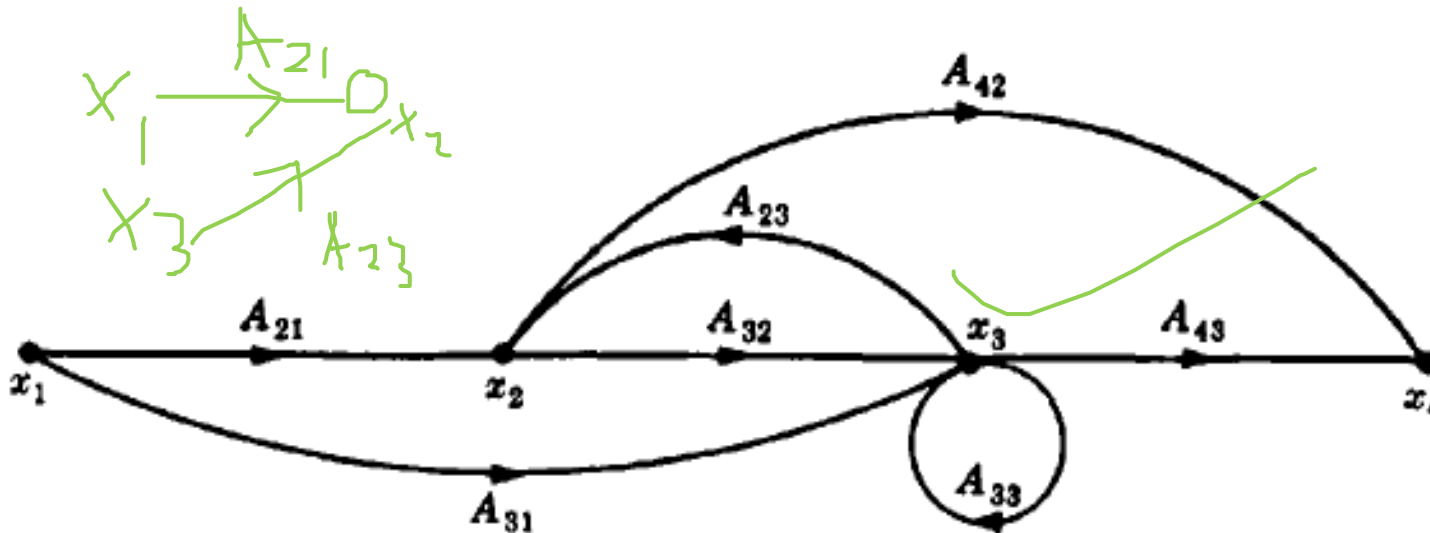
Construct the signal flow graph for the following set of simultaneous equations.

$$x_2 = A_{21}x_1 + A_{23}x_3$$

$$x_3 = A_{31}x_1 + A_{32}x_2 + A_{33}x_3$$

$$x_4 = A_{42}x_2 + A_{43}x_3$$

- There are four variables in the equations (i.e., x_1, x_2, x_3 , and x_4) therefore four nodes are required to construct the signal flow graph.
- Arrange these four nodes from left to right and connect them with the associated branches.



- Another way to arrange this graph is shown in the figure.

Terminologies

- An **input node** or source contain only the outgoing branches. i.e., X_1
- An **output node** or sink contain only the incoming branches. i.e., X_4
- A **path** is a continuous, unidirectional succession of branches along which no node is passed more than ones. i.e.,

X_1 to X_2 to X_3 to X_4

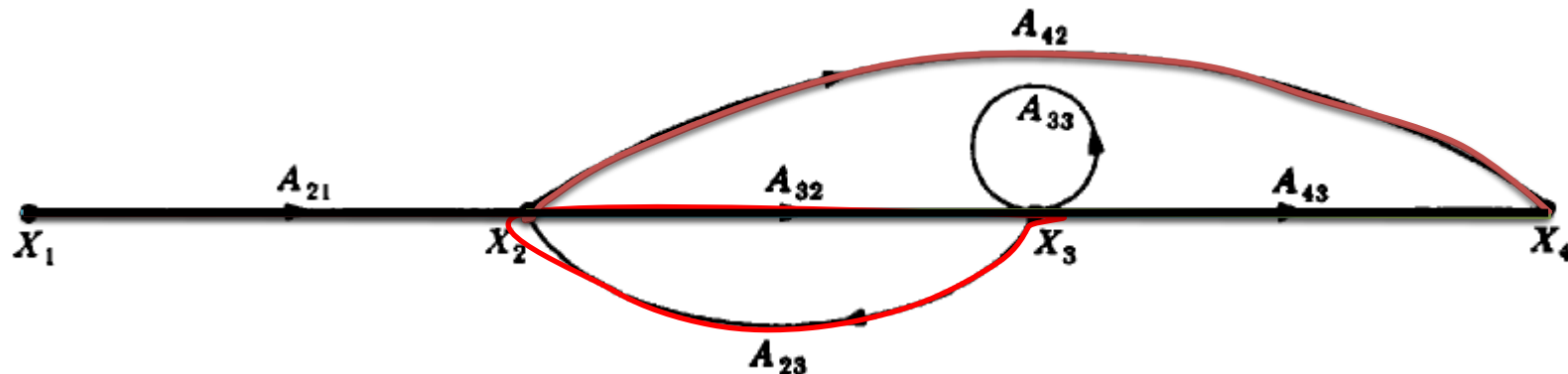
X_1 to X_2 to X_4

X_2 to X_3 to X_4

- A **forward path** is a path from the input node to the output node. i.e.,

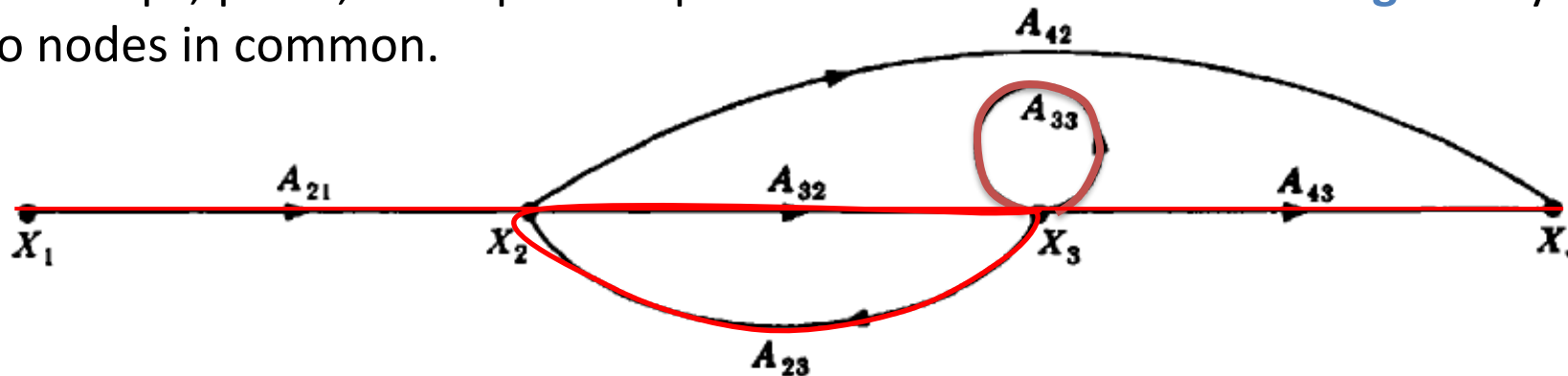
X_1 to X_2 to X_3 to X_4 , and X_1 to X_2 to X_4 , are forward paths.

- A **feedback path** or feedback loop is a path which originates and terminates on the same node. i.e.; X_2 to X_3 and back to X_2 is a feedback path.

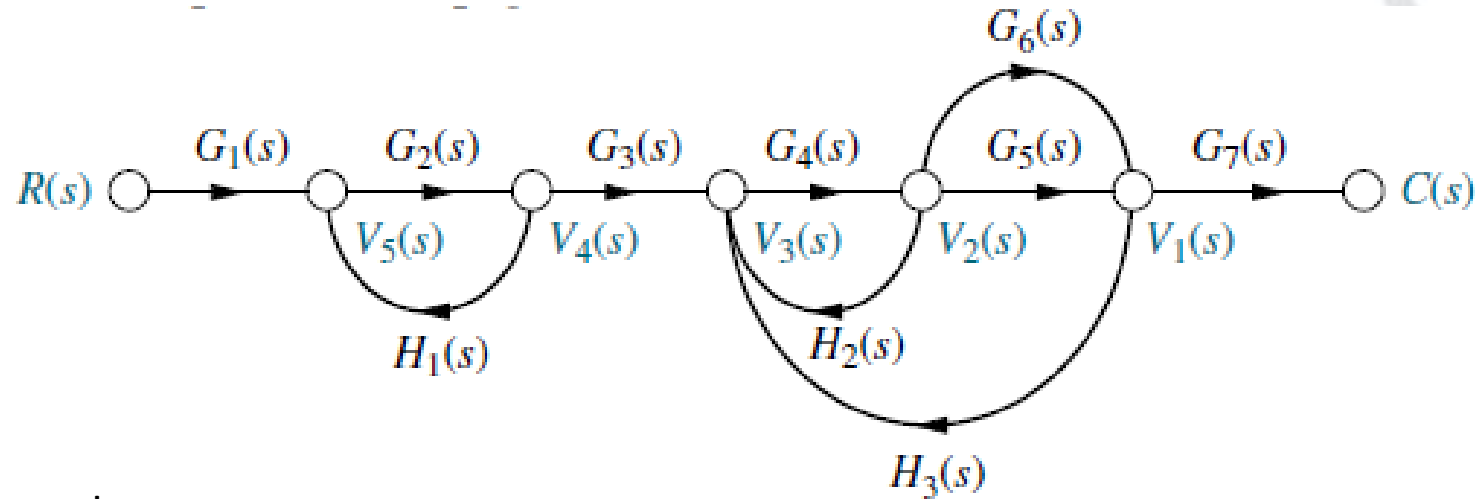


Terminologies

- A **self-loop** is a feedback loop consisting of a single branch. i.e.; A_{33} is a self loop.
- The **gain** of a branch is the transmission function of that branch.
- The **path gain** is the product of branch gains encountered in traversing a path.
i.e. the gain of forwards path X_1 to X_2 to X_3 to X_4 is $A_{21}A_{32}A_{43}$
- The **loop gain** is the product of the branch gains of the loop. i.e., the loop gain of the feedback loop from X_2 to X_3 and back to X_2 is $A_{32}A_{23}$.
- Two loops, paths, or loop and a path are said to be **non-touching** if they have no nodes in common.

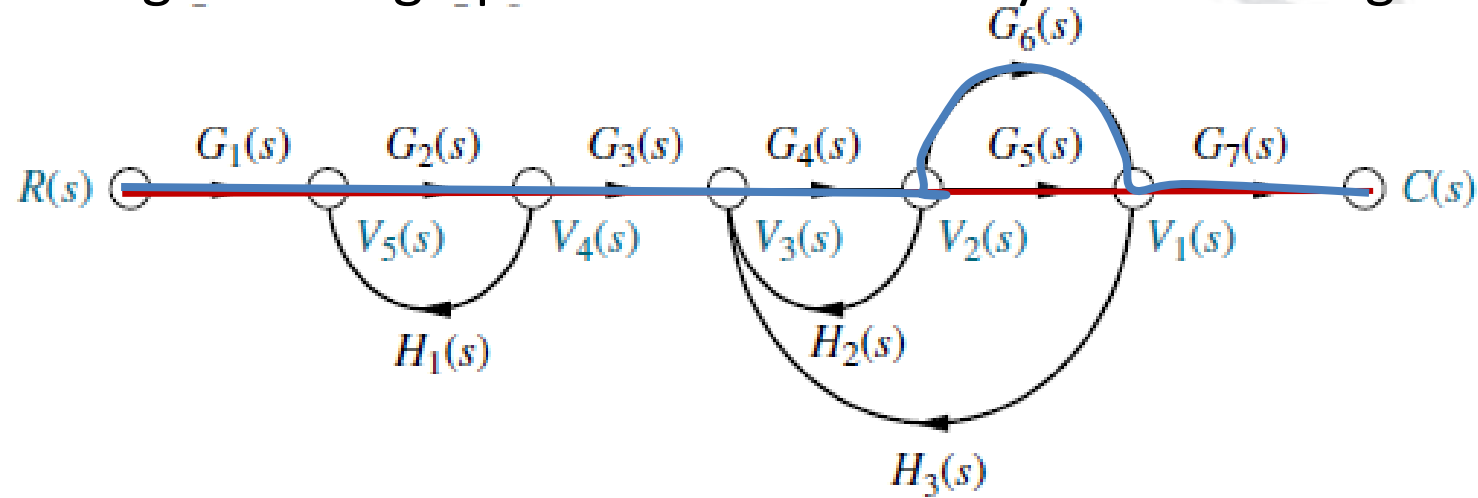


Consider the signal flow graph below and identify the following



- Input node.
- Output node.
- Forward paths.
- Feedback paths (loops).
- Determine the loop gains of the feedback loops.
- Determine the path gains of the forward paths.
- Non-touching loops

Consider the signal flow graph below and identify the following



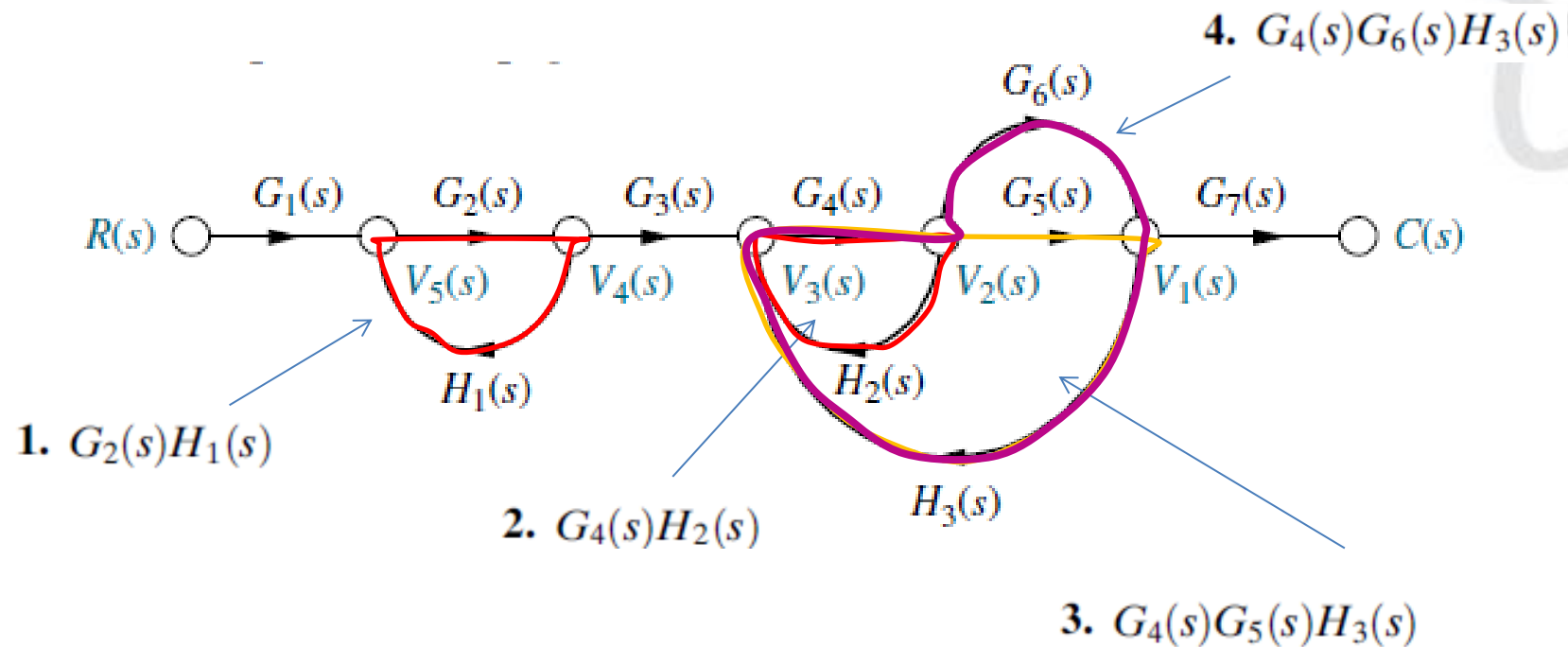
- There are two forward path gains;

1. $G_1(s)G_2(s)G_3(s)G_4(s)G_5(s)G_7(s)$

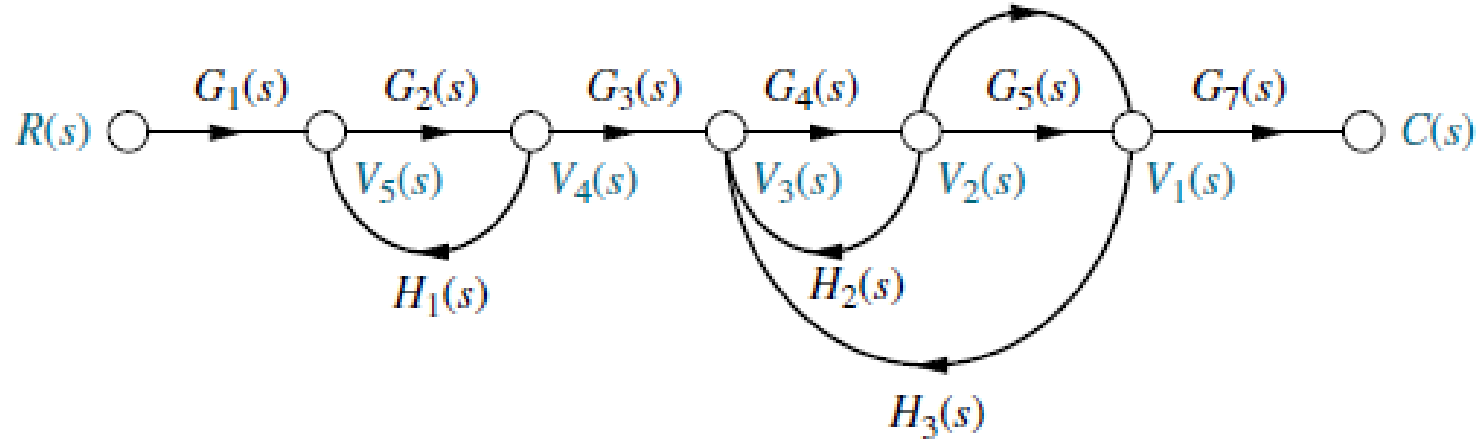
2. $G_1(s)G_2(s)G_3(s)G_4(s)G_6(s)G_7(s)$

Consider the signal flow graph below and identify the following

- There are four loops



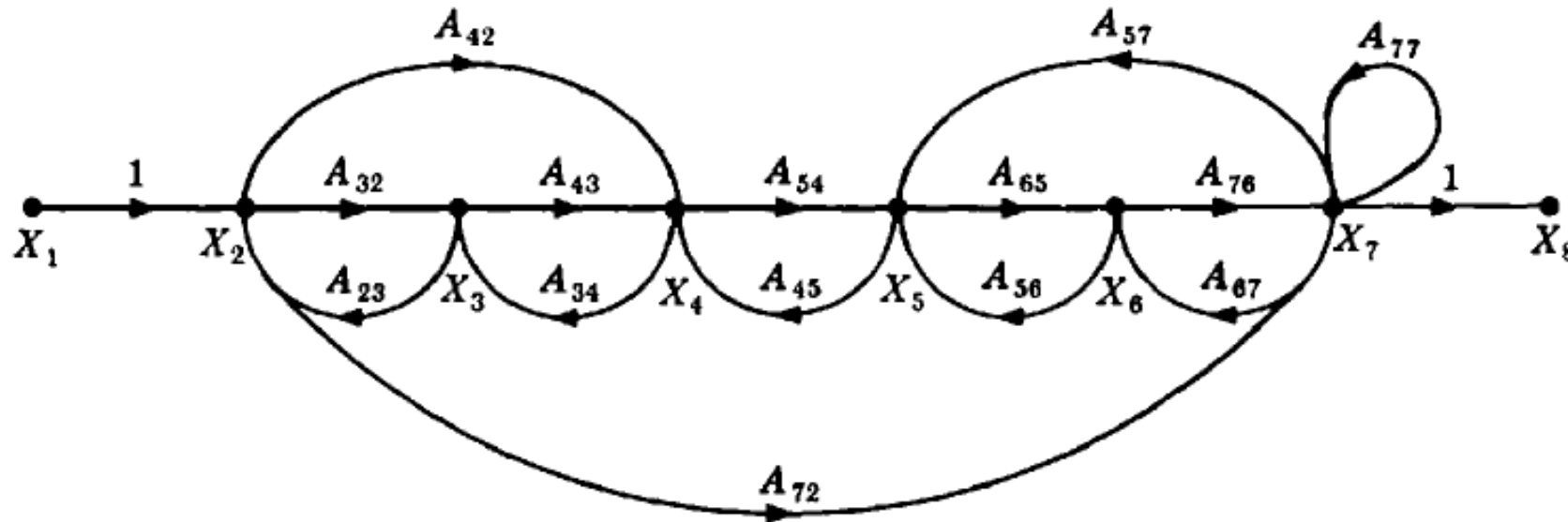
Consider the signal flow graph below and identify the following



- Nontouching loop gains;

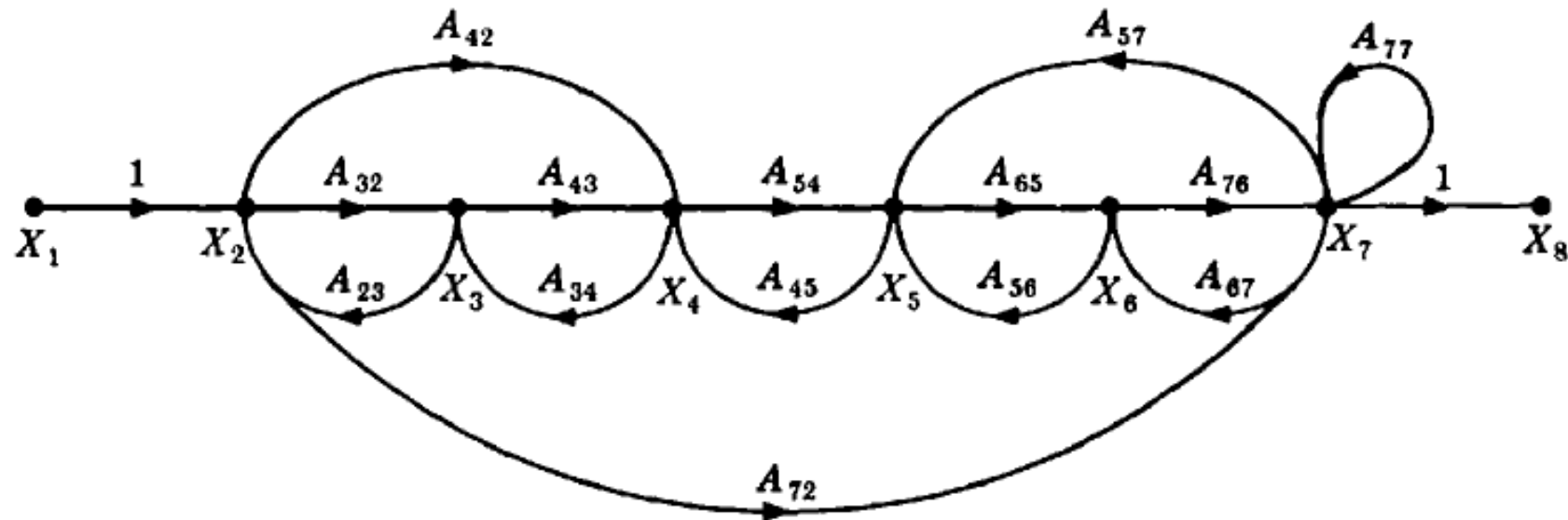
1. $[G_2(s)H_1(s)][G_4(s)H_2(s)]$
2. $[G_2(s)H_1(s)][G_4(s)G_5(s)H_3(s)]$
3. $[G_2(s)H_1(s)][G_4(s)G_6(s)H_3(s)]$

Consider the signal flow graph below and identify the following



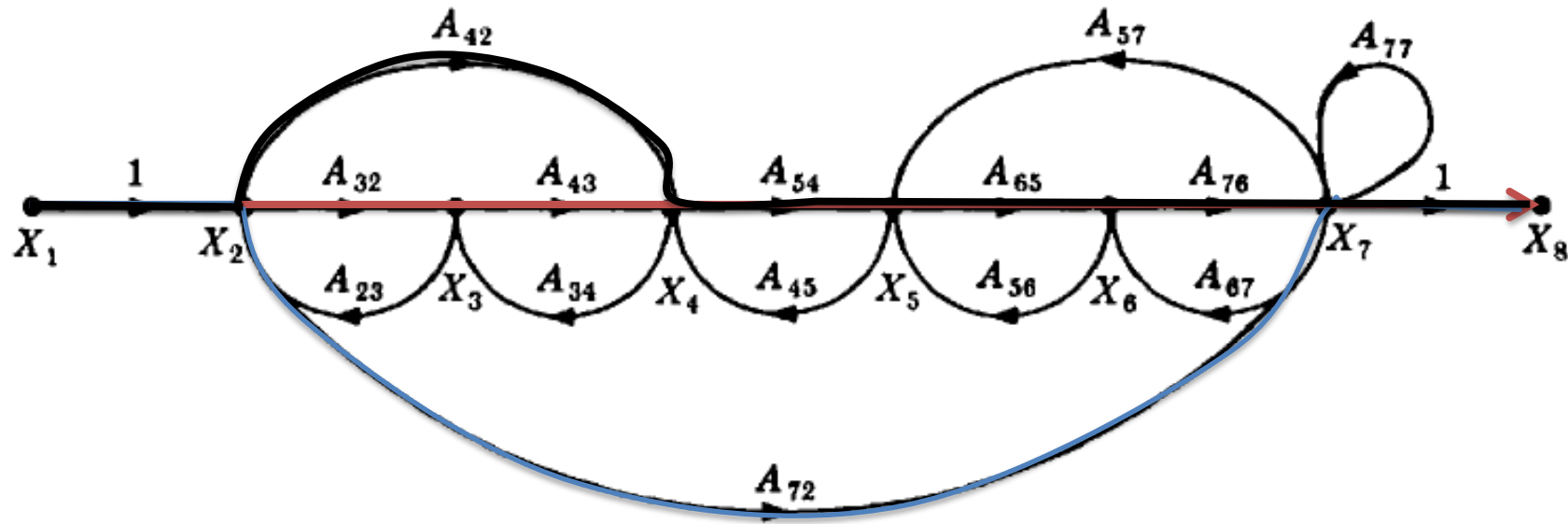
- Input node.
- Output node.
- Forward paths.
- Feedback paths.
- Self loop.
- Determine the loop gains of the feedback loops.
- Determine the path gains of the forward paths.

Input and output Nodes



- a) Input node X_1
- b) Output node X_8

(c) Forward Paths

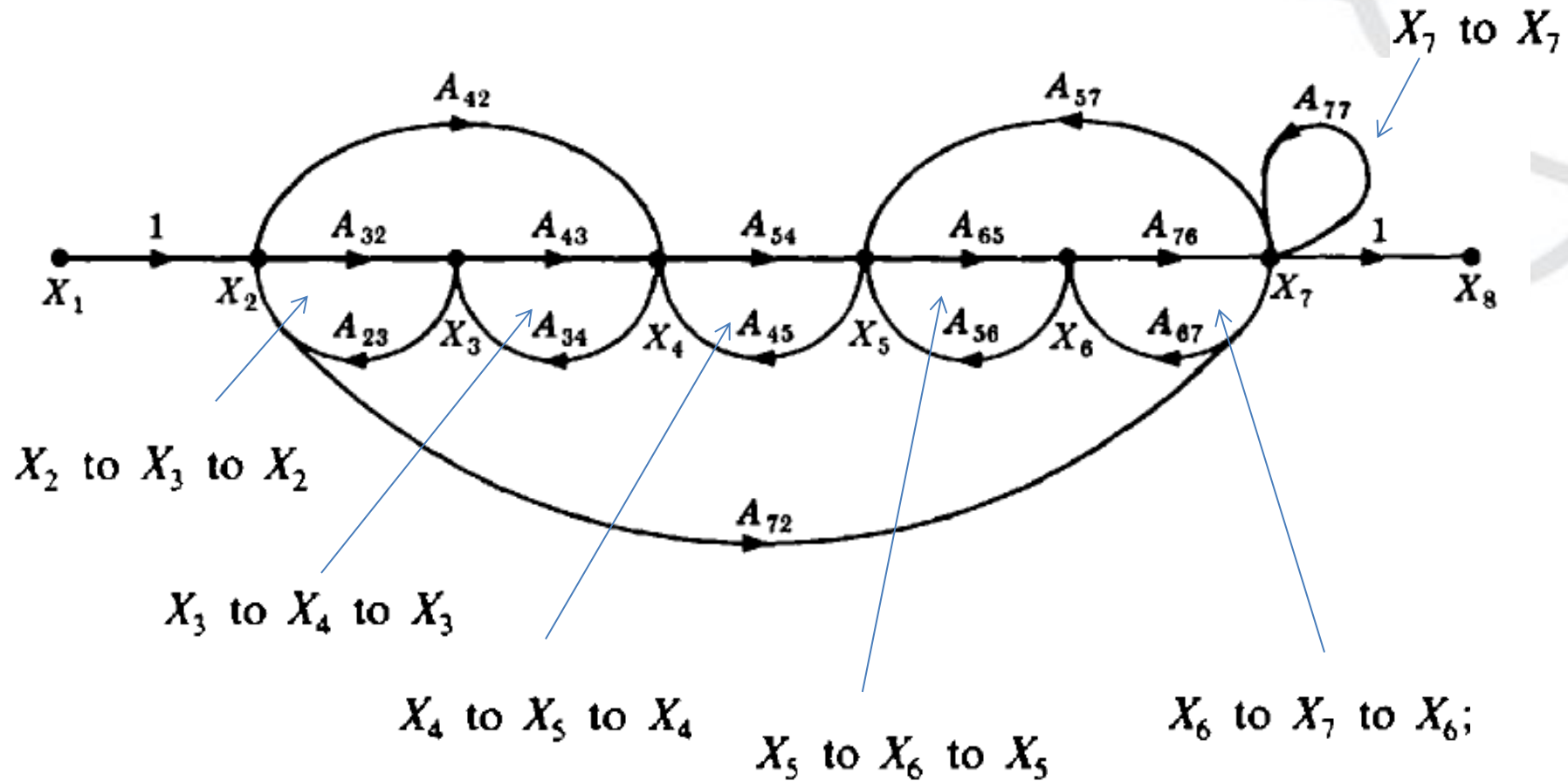


X_1 to X_2 to X_3 to X_4 to X_5 to X_6 to X_7 to X_8

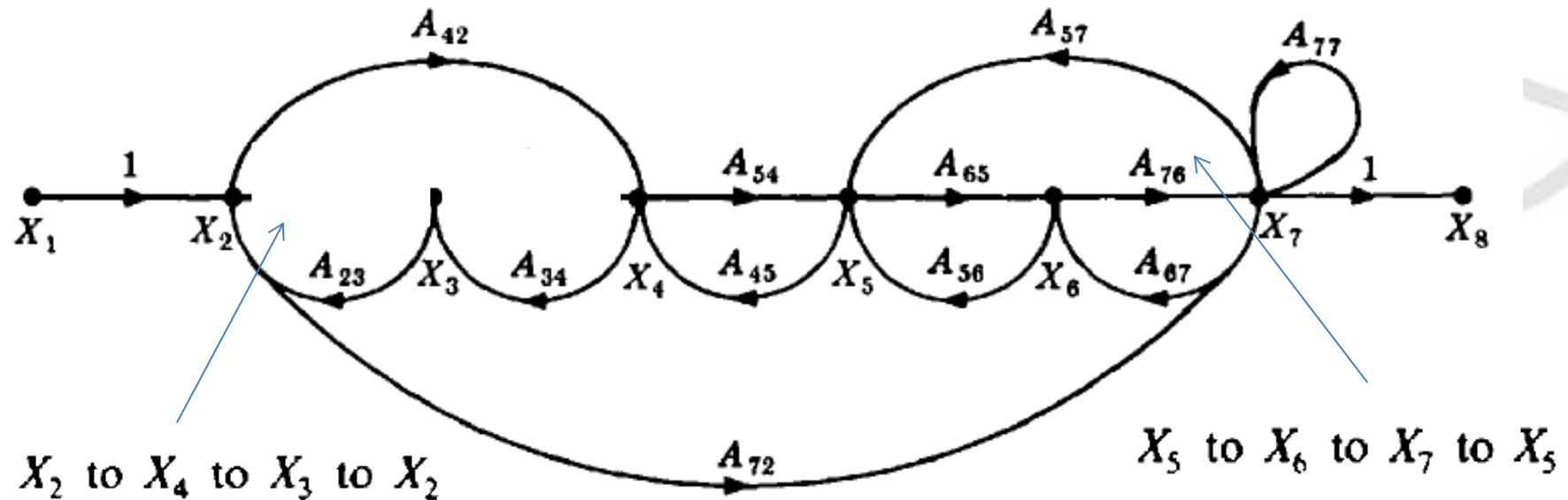
X_1 to X_2 to X_7 to X_8

X_1 to X_2 to X_4 to X_5 to X_6 to X_7 to X_8

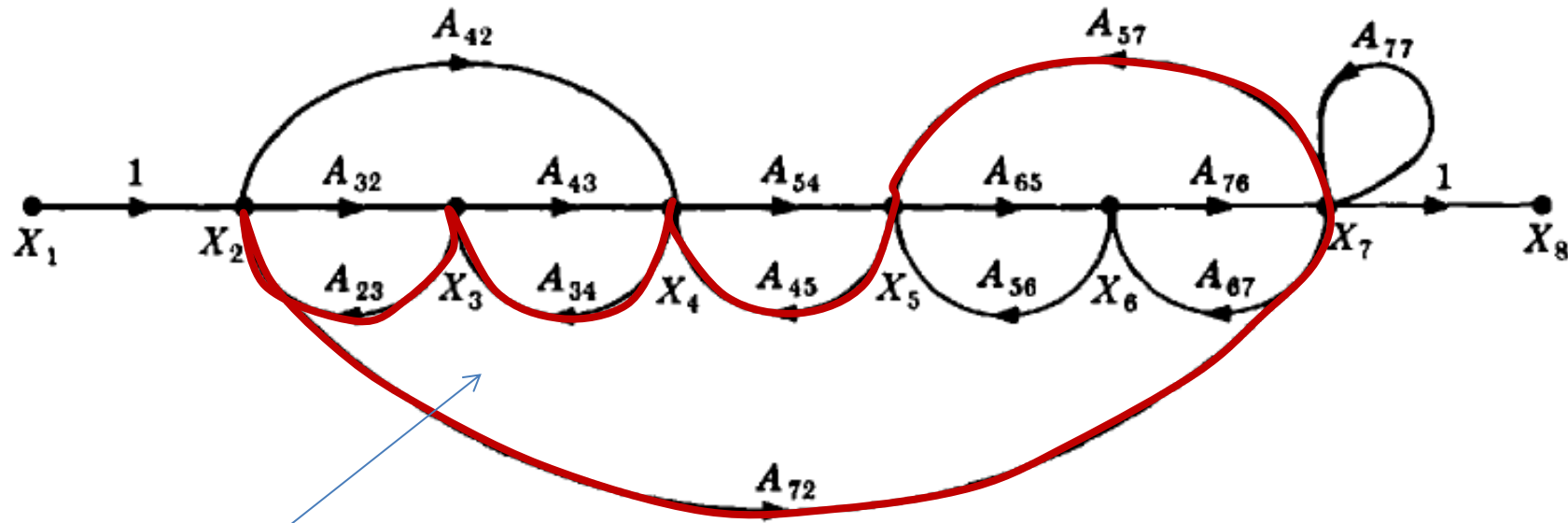
(d) Feedback Paths or Loops



(d) Feedback Paths or Loops

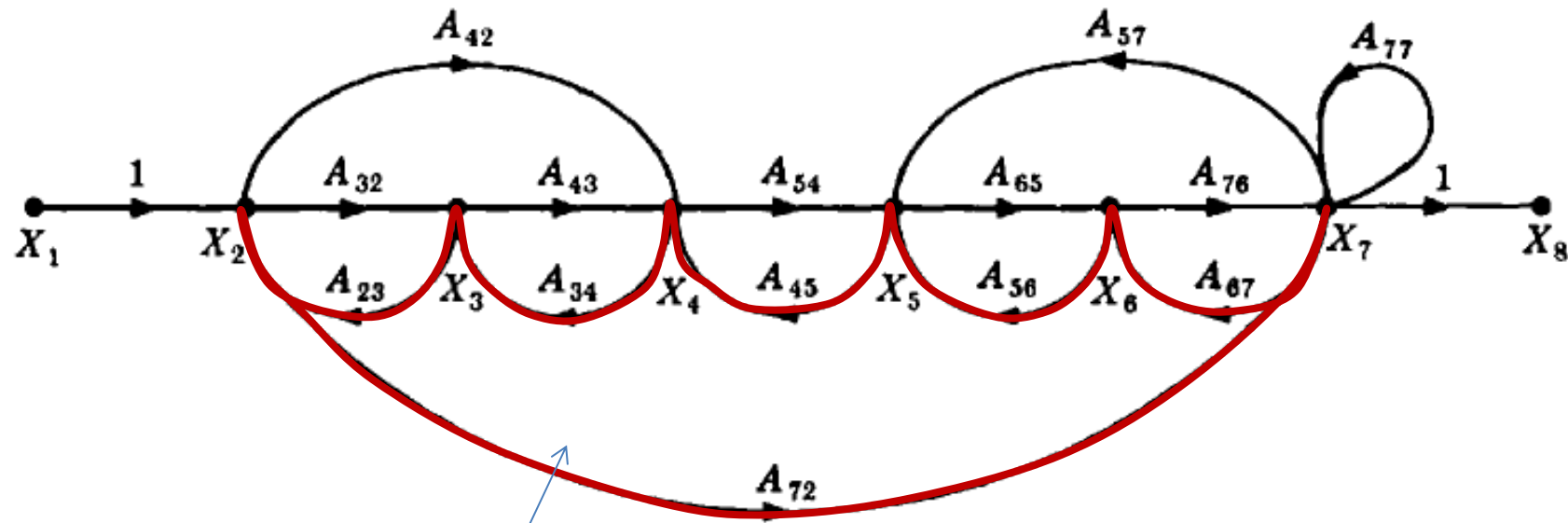


(d) Feedback Paths or Loops



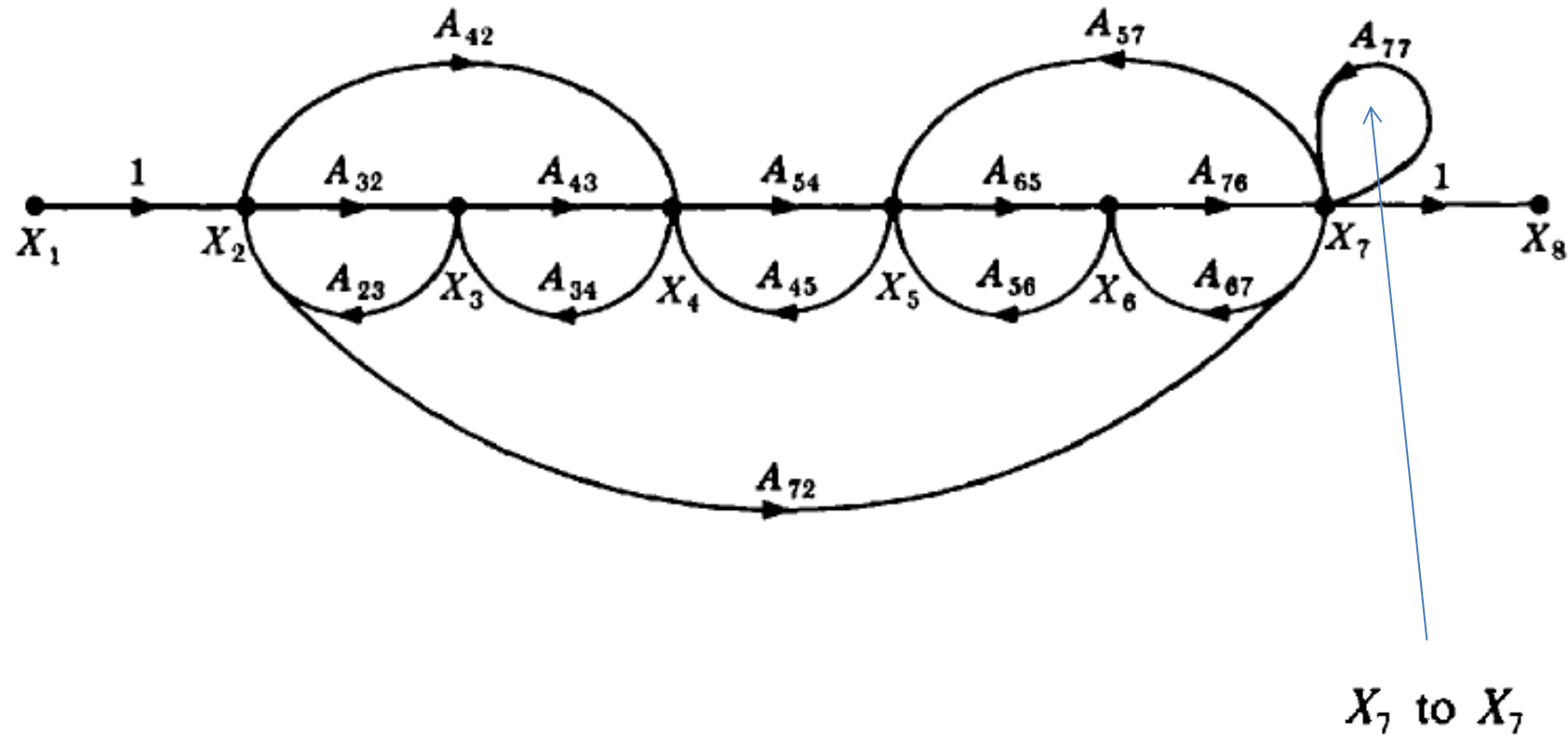
X_2 to X_7 to X_5 to X_4 to X_3 to X_2

(d) Feedback Paths or Loops

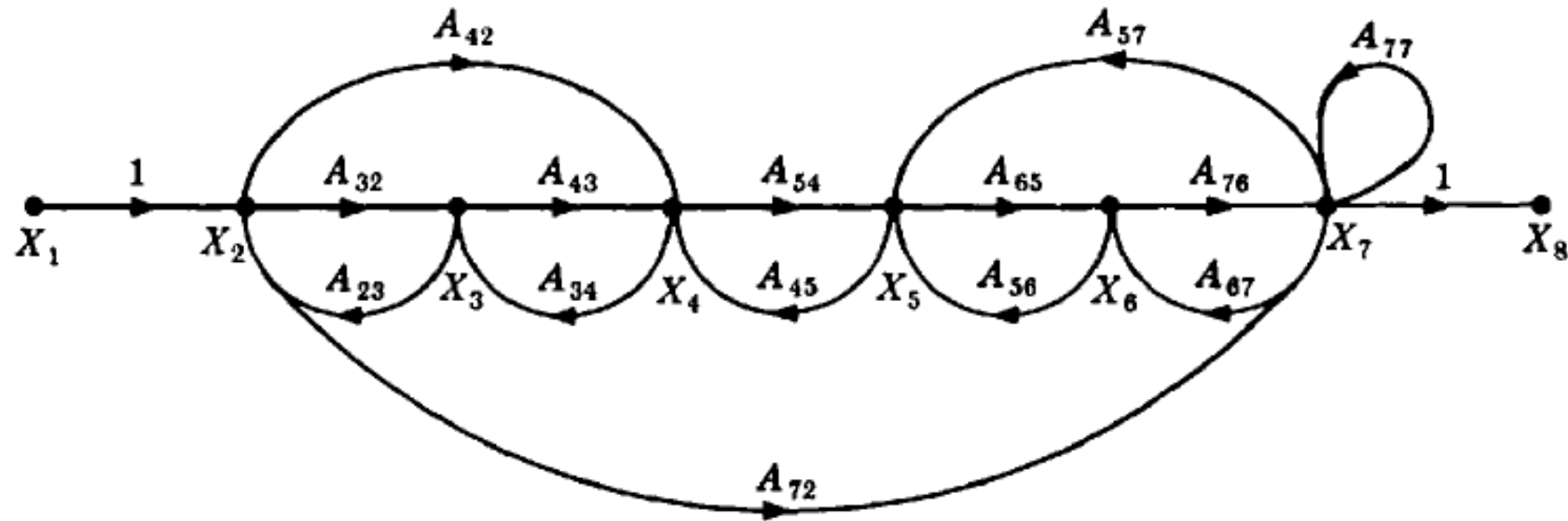


X_2 to X_7 to X_6 to X_5 to X_4 to X_3 to X_2

(e) Self Loop(s)



(f) Loop Gains of the Feedback Loops



$$A_{32} A_{23}$$

$$A_{76} A_{67}$$

$$A_{72} A_{57} A_{45} A_{34} A_{23}$$

$$A_{43} A_{34}$$

$$A_{65} A_{76} A_{57}$$

$$A_{72} A_{67} A_{56} A_{45} A_{34} A_{23}$$

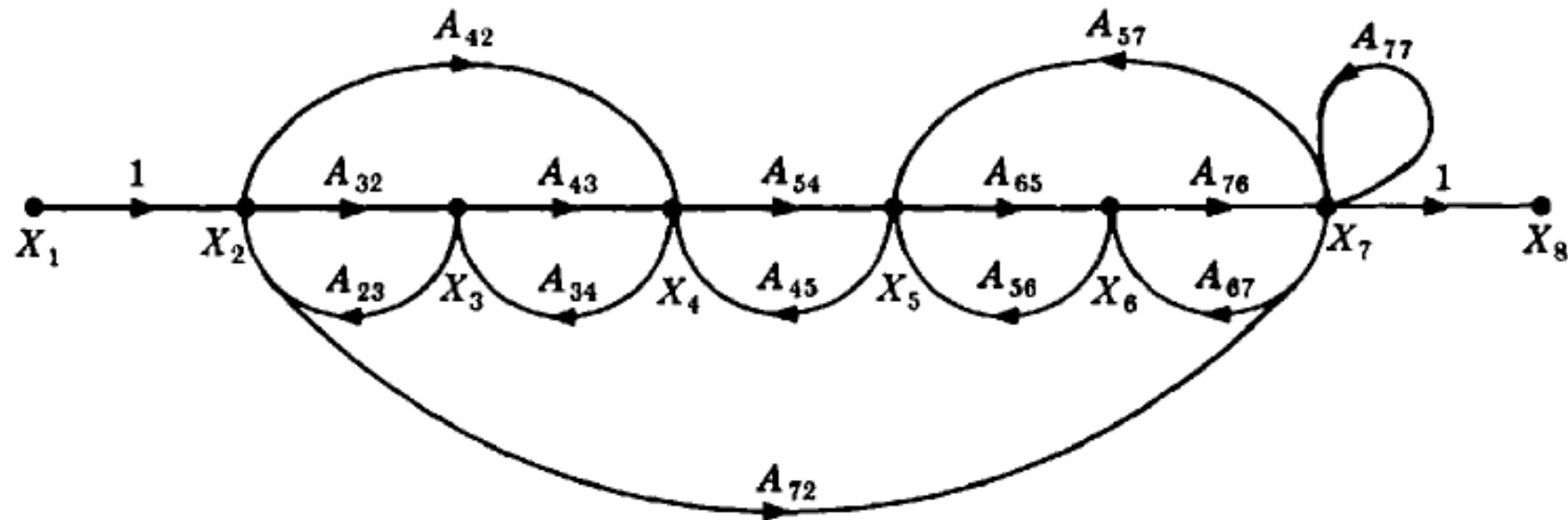
$$A_{54} A_{45}$$

$$A_{77}$$

$$A_{65} A_{56}$$

$$A_{42} A_{34} A_{23}$$

(g) Path Gains of the Forward Paths



$$A_{32} A_{43} A_{54} A_{65} A_{76}$$

$$A_{72}$$

$$A_{42} A_{54} A_{65} A_{76}$$

Mason's Rule (Mason, 1953)

- The block diagram reduction technique requires successive application of fundamental relationships in order to arrive at the system transfer function.
- On the other hand, Mason's rule for reducing a signal-flow graph to a single transfer function requires the application of one formula.
- The formula was derived by S. J. Mason when he related the signal-flow graph to the simultaneous equations that can be written from the graph.

Mason's Rule:

- The transfer function, $C(s)/R(s)$, of a system represented by a signal-flow graph is;

$$\frac{C(s)}{R(s)} = \frac{\sum_{i=1}^n P_i \Delta_i}{\Delta}$$

Where

n = number of forward paths.

P_i = the i^{th} forward-path gain.

Δ = Determinant of the system

Δ_i = Determinant of the i^{th} forward path

- Δ is called the signal flow graph determinant or characteristic function. Since $\Delta=0$ is the system characteristic equation.

Mason's Rule:

$$\frac{C(s)}{R(s)} = \frac{\sum_{i=1}^n P_i \Delta_i}{\Delta}$$

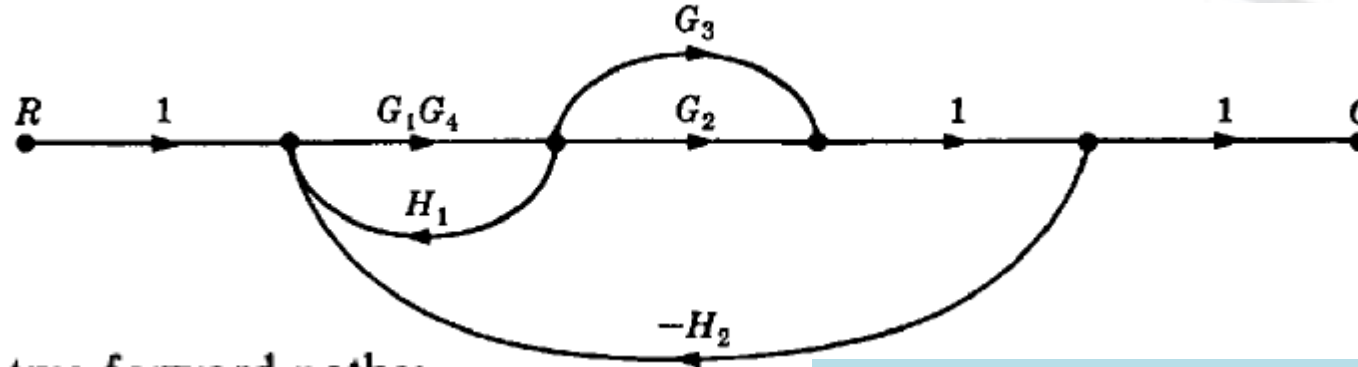
$\Delta = 1 -$ (sum of all individual loop gains) + (sum of the products of the gains of all possible two loops that do not touch each other) – (sum of the products of the gains of all possible three loops that do not touch each other) + ... and so forth with sums of higher number of non-touching loop gains

$\Delta_i =$ value of Δ for the part of the block diagram that does not touch the i -th forward path ($\Delta_i = 1$ if there are no non-touching loops to the i -th path.)

Systematic approach

1. Calculate forward path gain P_i for each forward path i .
2. Calculate all loop transfer functions
3. Consider non-touching loops 2 at a time
4. Consider non-touching loops 3 at a time
5. etc
6. Calculate Δ from steps 2,3,4 and 5
7. Calculate Δ_i as portion of Δ not touching forward path i

Example#1: Apply Mason's Rule to calculate the transfer function of the system represented by following Signal Flow Graph



There are two forward paths:

$$P_1 = G_1G_2G_4 \quad P_2 = G_1G_3G_4$$

Calculate forward path gain P_i for each forward path i .

Therefore,

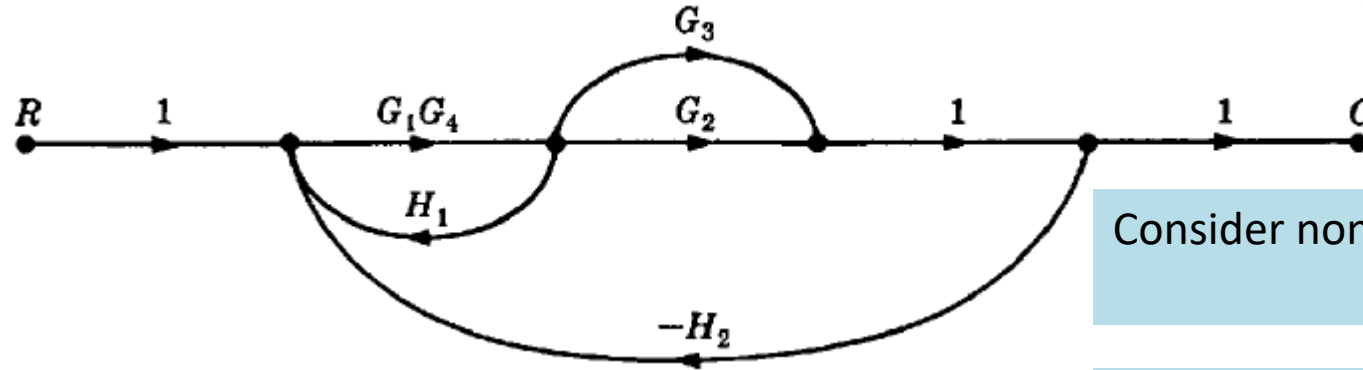
$$\frac{C}{R} = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta}$$

There are three feedback loops

Calculate all loop transfer functions

$$L_1 = G_1G_4H_1, \quad L_2 = -G_1G_2G_4H_2, \quad L_3 = -G_1G_3G_4H_2$$

Example#1: Apply Mason's Rule to calculate the transfer function of the system represented by following Signal Flow Graph



Consider non-touching loops 2 at a time

Consider non-touching loops 3 at a time

There are no non-touching loops, therefore

$$\Delta = 1 - (\text{sum of all individual loop gains})$$

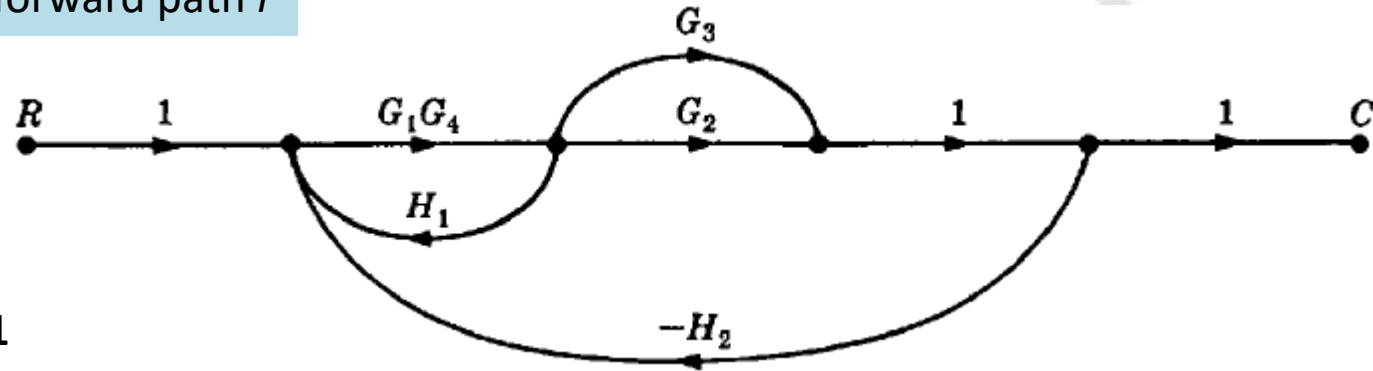
Calculate Δ from steps 2,3,4 and 5

$$\Delta = 1 - (L_1 + L_2 + L_3)$$

$$\Delta = 1 - (G_1G_4H_1 - G_1G_2G_4H_2 - G_1G_3G_4H_2)$$

Example#1: Apply Mason's Rule to calculate the transfer function of the system represented by following Signal Flow Graph

Calculate Δ_i as portion of Δ not touching forward path i



Eliminate forward path-1

$$\Delta_1 = 1 - (\text{sum of all individual loop gains}) + \dots$$

$$\Delta_1 = 1$$

Eliminate forward path-2

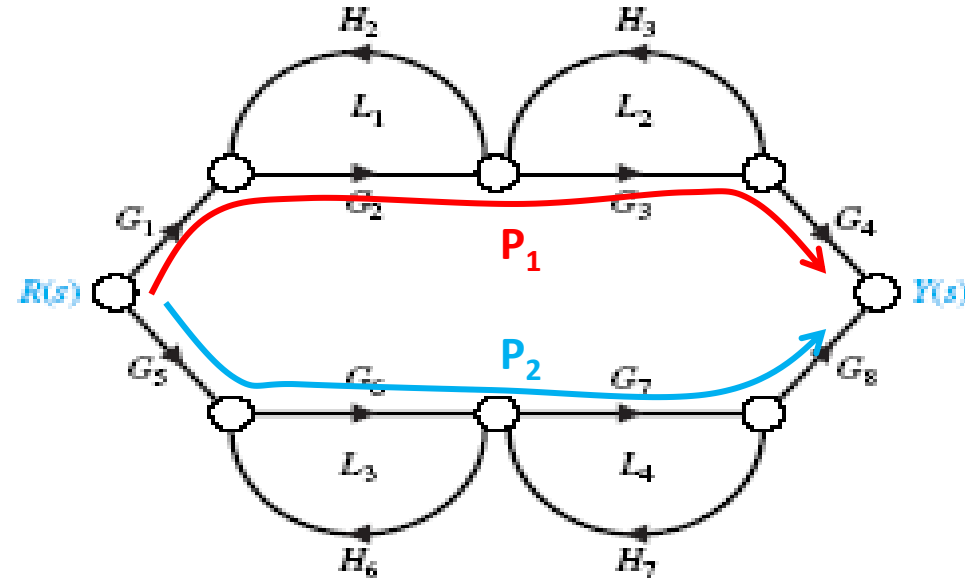
$$\Delta_2 = 1 - (\text{sum of all individual loop gains}) + \dots$$

$$\Delta_2 = 1$$

Example#1: Continue

$$\begin{aligned}\frac{C}{R} &= \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta} = \frac{G_1G_2G_4 + G_1G_3G_4}{1 - G_1G_4H_1 + G_1G_2G_4H_2 + G_1G_3G_4H_2} \\ &= \frac{G_1G_4(G_2 + G_3)}{1 - G_1G_4H_1 + G_1G_2G_4H_2 + G_1G_3G_4H_2}\end{aligned}$$

Example#2: Apply Mason's Rule to calculate the transfer function of the system represented by following Signal Flow Graph



1. Calculate forward path gains for each forward path.

$$P_1 = G_1 G_2 G_3 G_4 \text{ (path 1)} \quad \text{and} \quad P_2 = G_5 G_6 G_7 G_8 \text{ (path 2)}$$

2. Calculate all loop gains.

$$L_1 = G_2 H_2, \quad L_2 = H_3 G_3, \quad L_3 = G_6 H_6, \quad L_4 = G_7 H_7$$

3. Consider two non-touching loops.

$$\begin{matrix} L_1 L_3 & L_1 L_4 \\ L_2 L_4 & L_2 L_3 \end{matrix}$$

Example#2: continue

4. Consider three non-touching loops.

None.

5. Calculate Δ from steps 2,3,4.

$$\Delta = 1 - (L_1 + L_2 + L_3 + L_4) + (L_1L_3 + L_1L_4 + L_2L_3 + L_2L_4)$$

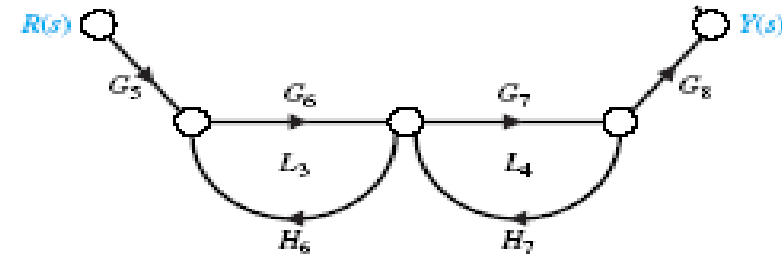
$$\Delta = 1 - (G_2H_2 + H_3G_3 + G_6H_6 + G_7H_7) + \\ (G_2H_2G_6H_6 + G_2H_2G_7H_7 + H_3G_3G_6H_6 + H_3G_3G_7H_7)$$

Example#2: continue

Eliminate forward path-1

$$\Delta_1 = 1 - (L_3 + L_4)$$

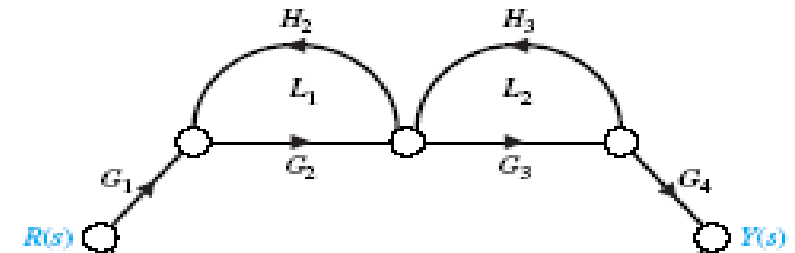
$$\Delta_1 = 1 - (G_6 H_6 + G_7 H_7)$$



Eliminate forward path-2

$$\Delta_2 = 1 - (L_1 + L_2)$$

$$\Delta_2 = 1 - (G_2 H_2 + G_3 H_3)$$



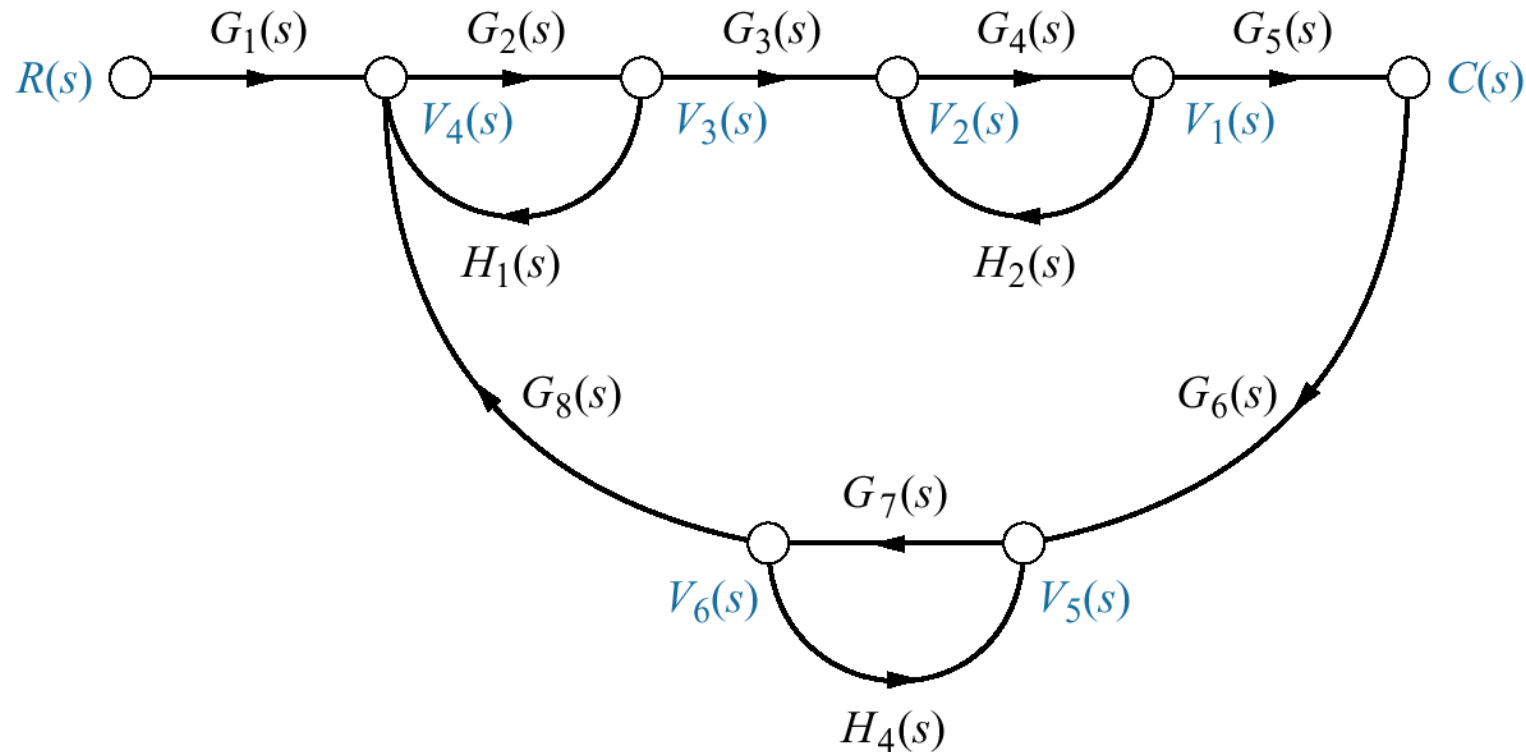
Example#2: continue

$$\frac{Y(s)}{R(s)} = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta}$$

$$\frac{Y(s)}{R(s)} = \frac{G_1G_2G_3G_4[1 - (G_6H_6 + G_7H_7)] + G_5G_6G_7G_8[1 - (G_2H_2 + G_3H_3)]}{1 - (G_2H_2 + H_3G_3 + G_6H_6 + G_7H_7) + (G_2H_2G_6H_6 + G_2H_2G_7H_7 + H_3G_3G_6H_6 + H_3G_3G_7H_7)}$$

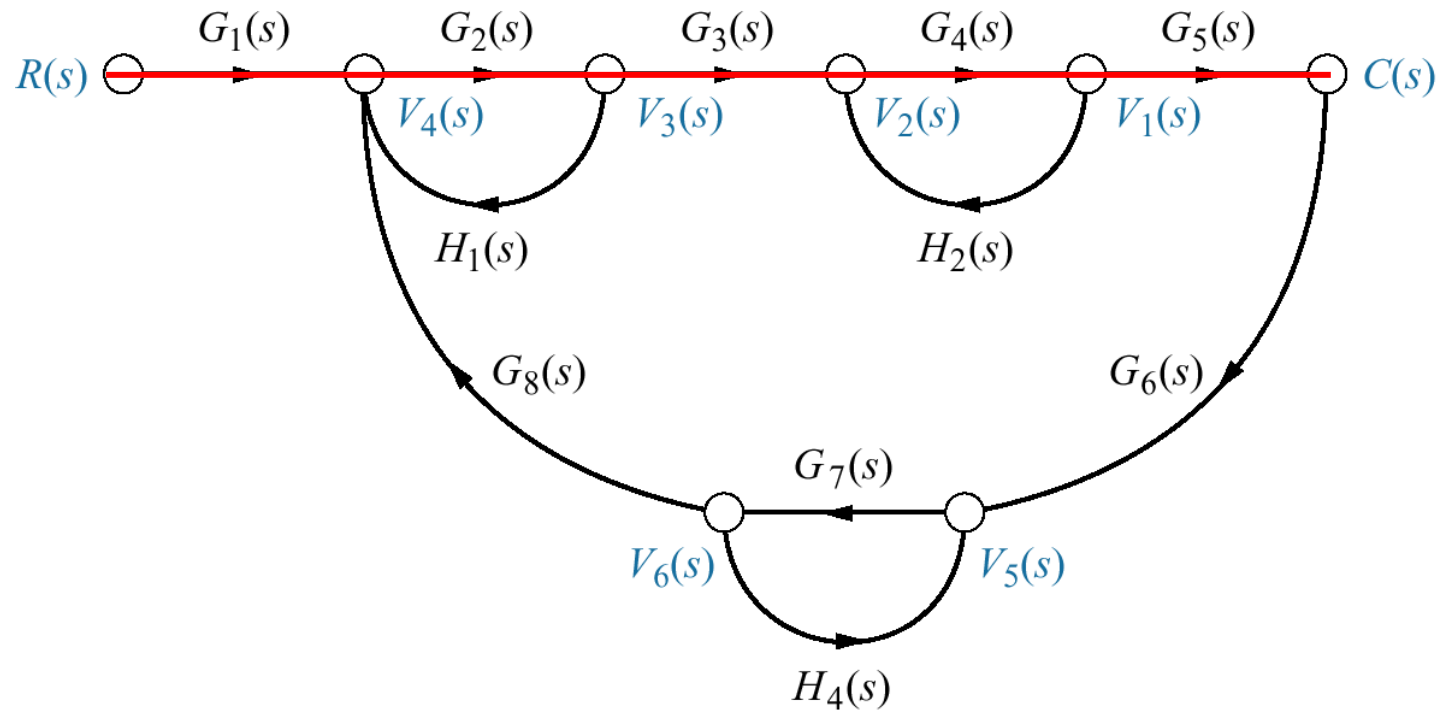
Example#3

- Find the transfer function, $C(s)/R(s)$, for the signal-flow graph in figure below.



Example#3

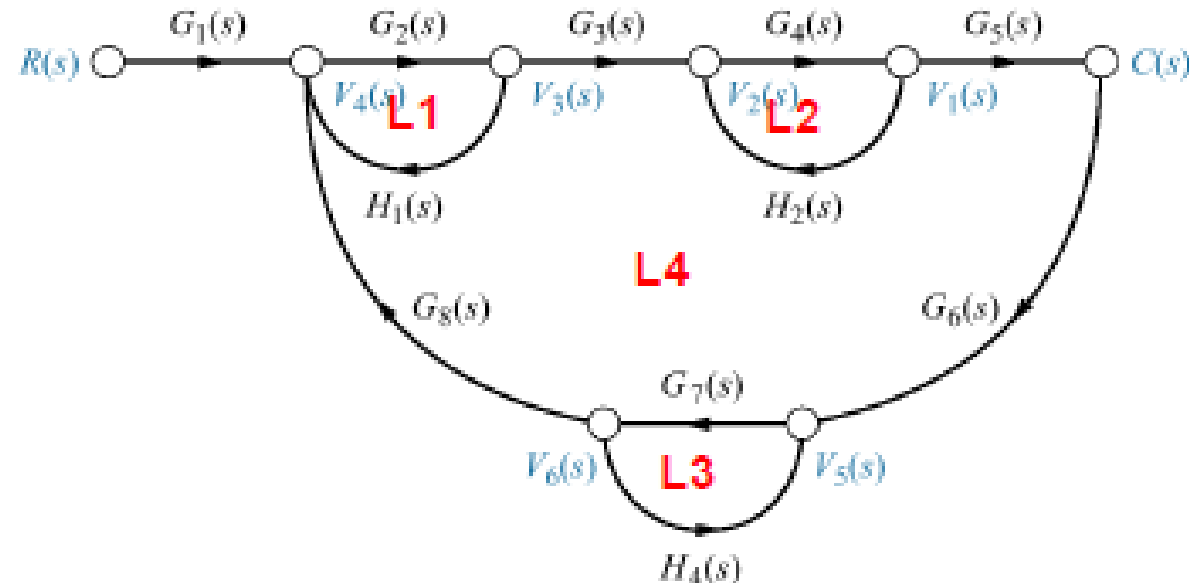
- There is only one forward Path.



$$P_1 = G_1(s)G_2(s)G_3(s)G_4(s)G_5(s)$$

Example#3

- There are four feedback loops.



L1. $G_2(s)H_1(s)$

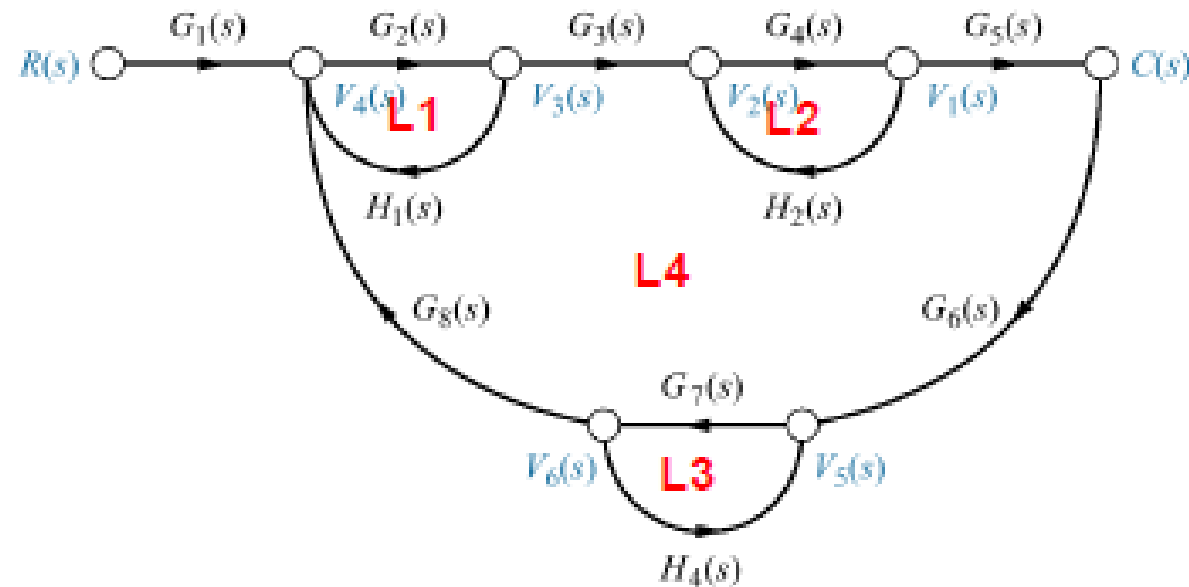
L3. $G_7(s)H_4(s)$

L2. $G_4(s)H_2(s)$

L4. $G_2(s)G_3(s)G_4(s)G_5(s)G_6(s)G_7(s)G_8(s)$

Example#3

- Non-touching loops taken two at a time.

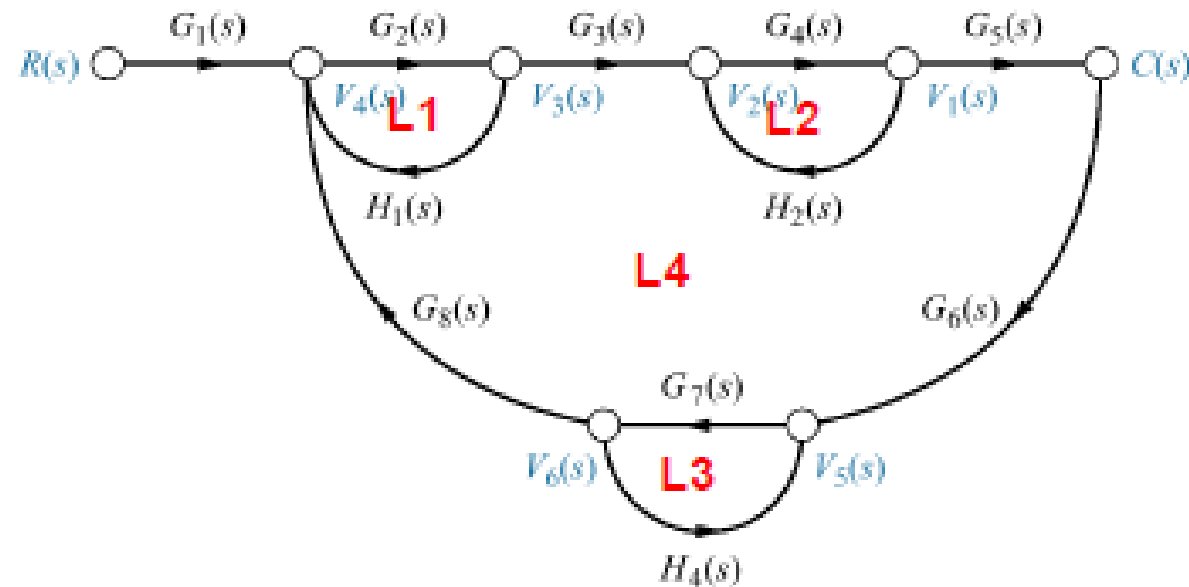


L1 and L2: $G_2(s)H_1(s)G_4(s)H_2(s)$ L2 and L3: $G_4(s)H_2(s)G_7(s)H_4(s)$

L1 and L3: $G_2(s)H_1(s)G_7(s)H_4(s)$

Example#3

- Non-touching loops taken three at a time.



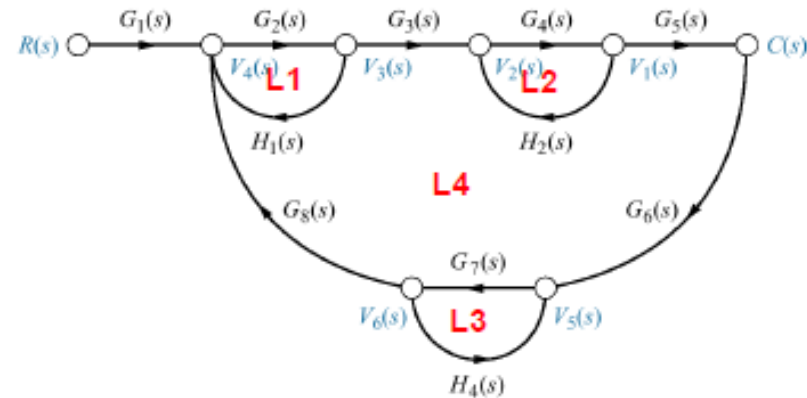
L1, L2, L3: $G_2(s)H_1(s)G_4(s)H_2(s)G_7(s)H_4(s)$

Example#3

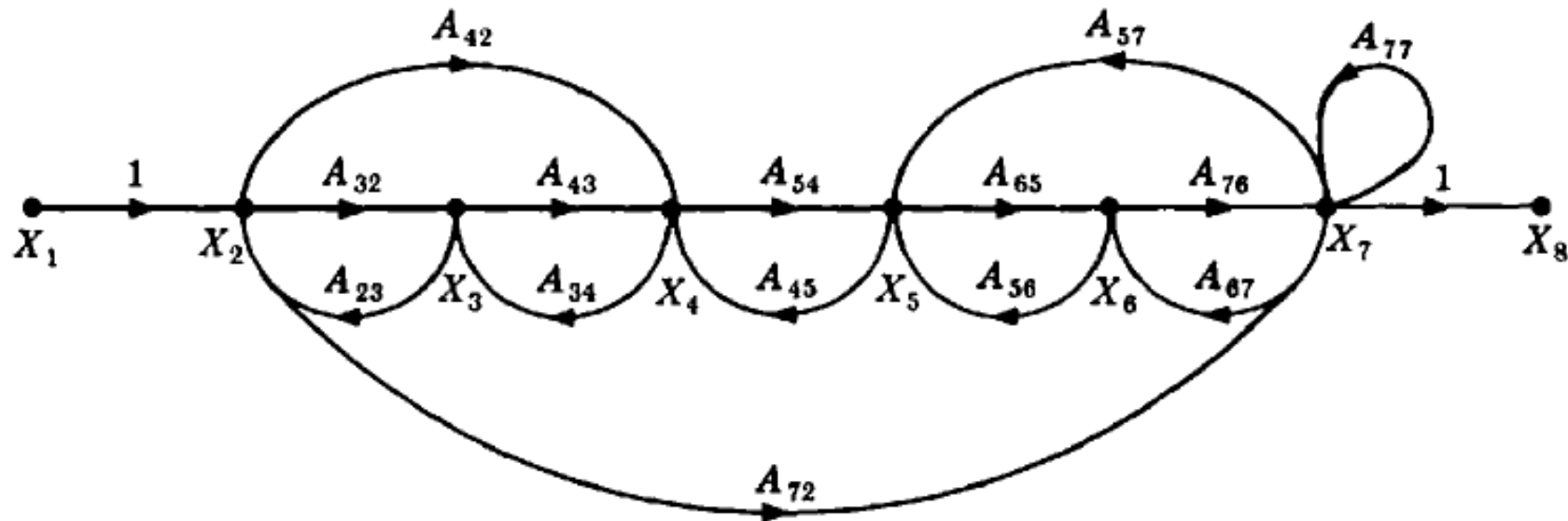
$$\begin{aligned}\Delta = & 1 - [G_2(s)H_1(s) + G_4(s)H_2(s) \\ & + G_7(s)H_4(s) + G_2(s)G_3(s)G_4(s)G_5(s)G_6(s)G_7(s)G_8(s)] \\ & + [G_2(s)H_1(s)G_4(s)H_2(s) + G_2(s)H_1(s)G_7(s)H_4(s) \\ & + G_4(s)H_2(s)G_7(s)H_4(s)] \\ & - [G_2(s)H_1(s)G_4(s)H_2(s)G_7(s)H_4(s)]\end{aligned}$$

Eliminate forward path-1

$$\Delta_1 = 1 - G_7(s)H_4(s)$$



Example#4: Apply Mason's Rule to calculate the transfer function of the system represented by following Signal Flow Graph

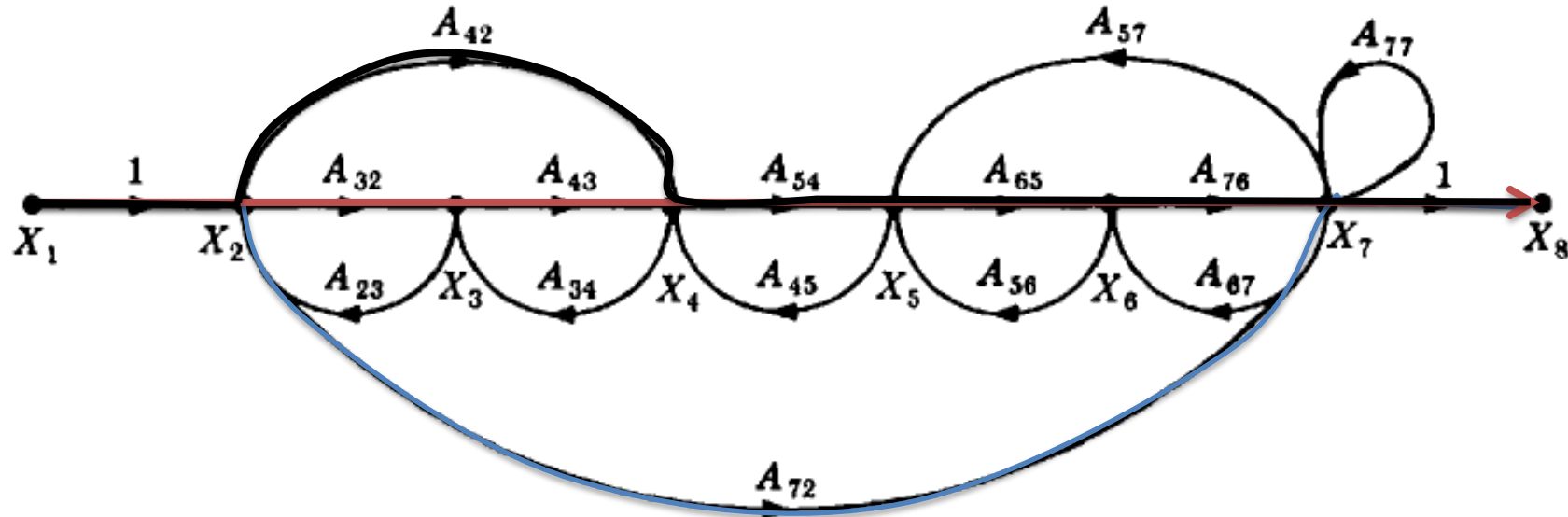


There are three forward paths, therefore $n=3$.

$$\frac{C(s)}{R(s)} = \frac{\sum_{i=1}^3 P_i \Delta_i}{\Delta} = \frac{P_1 \Delta_1 + P_2 \Delta_2 + P_3 \Delta_3}{\Delta}$$

Example#4: Forward Paths

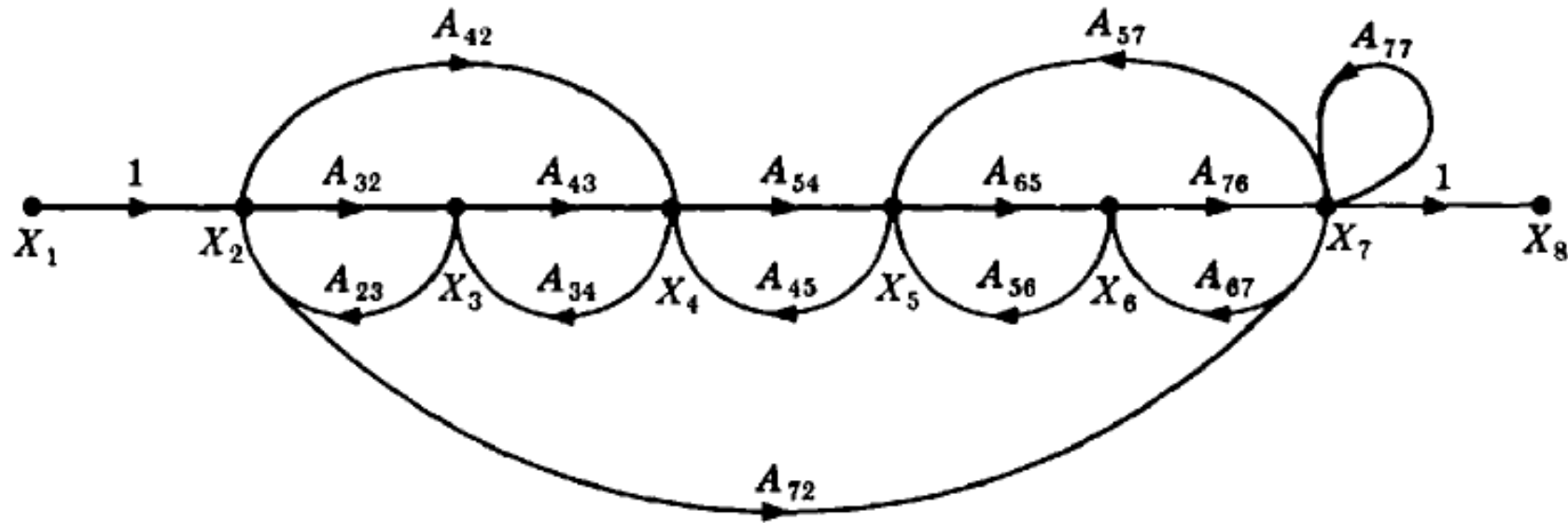
$$P_3 = A_{42} A_{54} A_{65} A_{76}$$



$$P_1 = A_{32} A_{43} A_{54} A_{65} A_{76}$$

$$P_2 = A_{72}$$

Example#4: Loop Gains of the Feedback Loops



$$L_1 = A_{32}A_{23}$$

$$L_2 = A_{43}A_{34}$$

$$L_3 = A_{54}A_{45}$$

$$L_4 = A_{65}A_{56}$$

$$L_5 = A_{76}A_{67}$$

$$L_6 = A_{77}$$

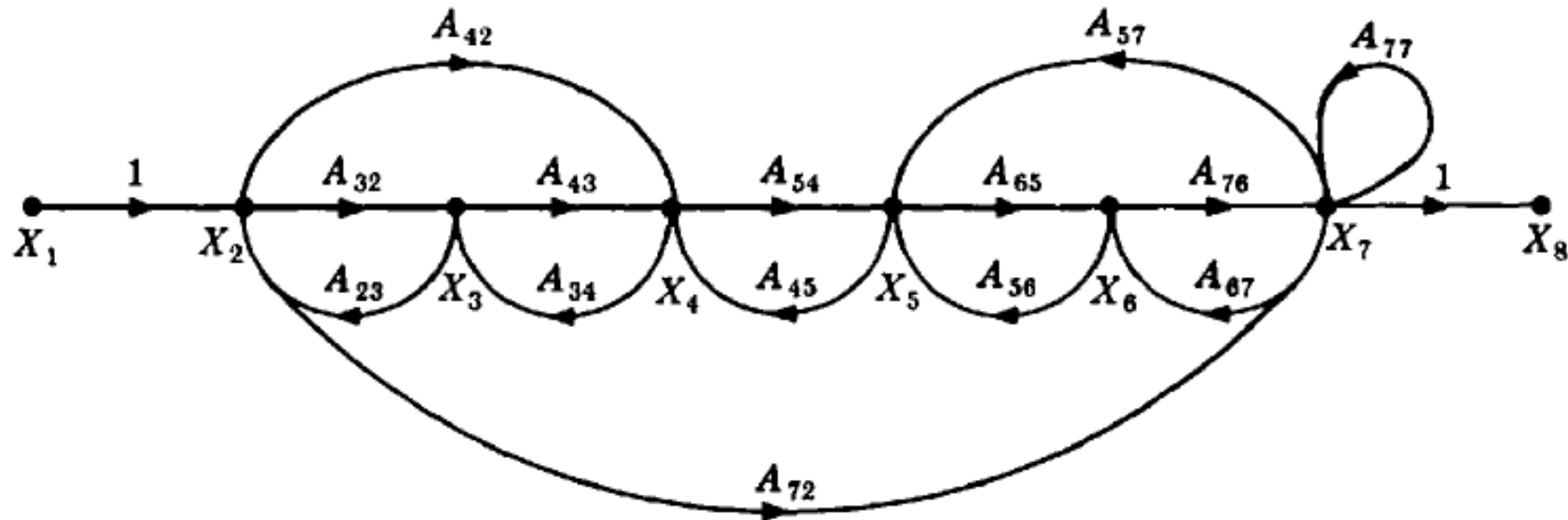
$$L_7 = A_{42}A_{34}A_{23}$$

$$L_8 = A_{65}A_{76}A_{67}$$

$$L_9 = A_{72}A_{57}A_{45}A_{34}A_{23}$$

$$L_{10} = A_{72}A_{67}A_{56}A_{45}A_{34}A_{23}$$

Example#4: two non-touching loops



$L_1 L_3$

$L_2 L_4$

$L_3 L_5$

$L_4 L_6$

$L_5 L_7$

$L_7 L_8$

$L_1 L_4$

$L_2 L_5$

$L_3 L_6$

$L_4 L_7$

$L_1 L_5$

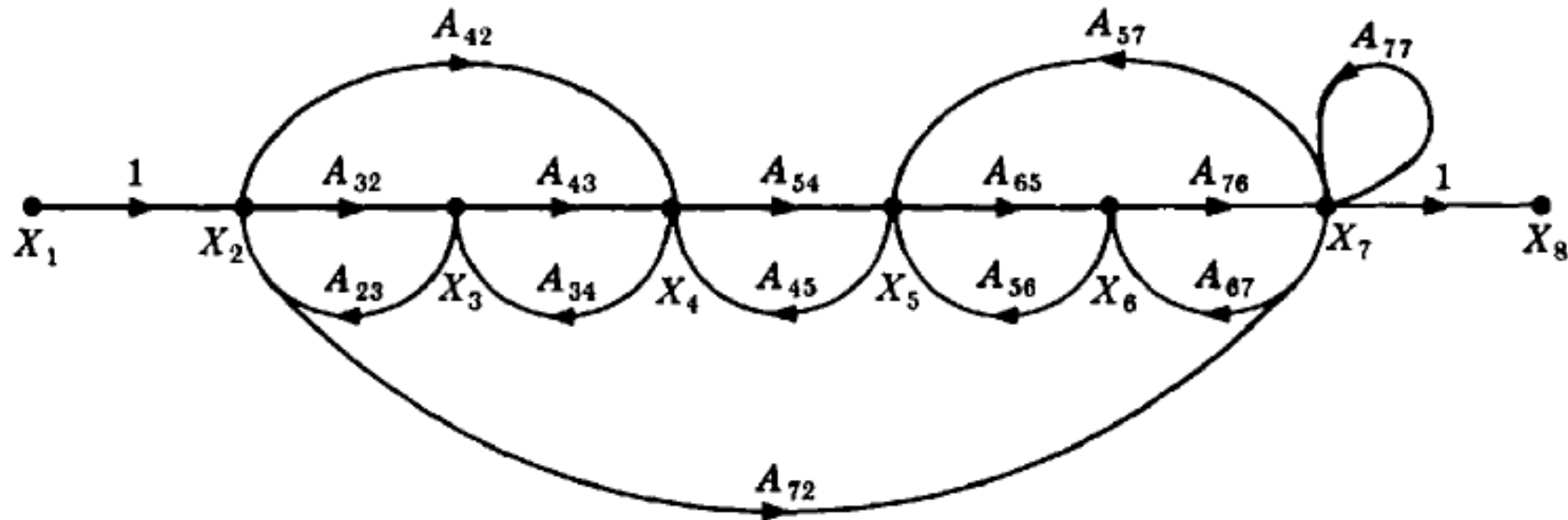
$L_2 L_6$

$L_1 L_6$

$L_2 L_8$

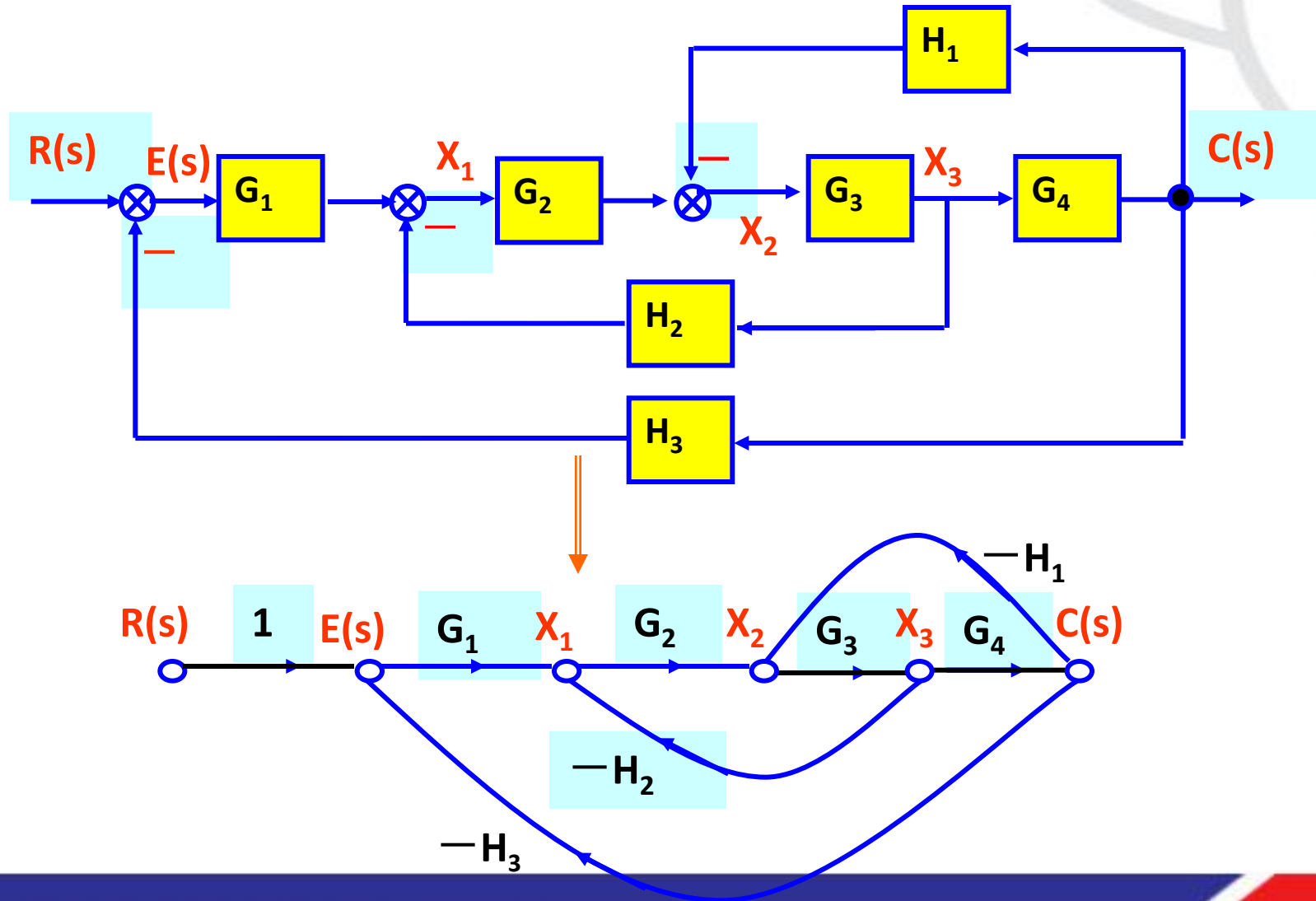
$L_1 L_8$

Example#4: Three non-touching loops


 $L_1 L_3$
 $L_2 L_4$
 $L_3 L_5$
 $L_4 L_6$
 $L_5 L_7$
 $L_7 L_8$
 $L_1 L_4$
 $L_2 L_5$
 $L_3 L_6$
 $L_4 L_7$
 $L_1 L_5$
 $L_2 L_6$
 $L_1 L_6$
 $L_2 L_8$
 $L_1 L_8$

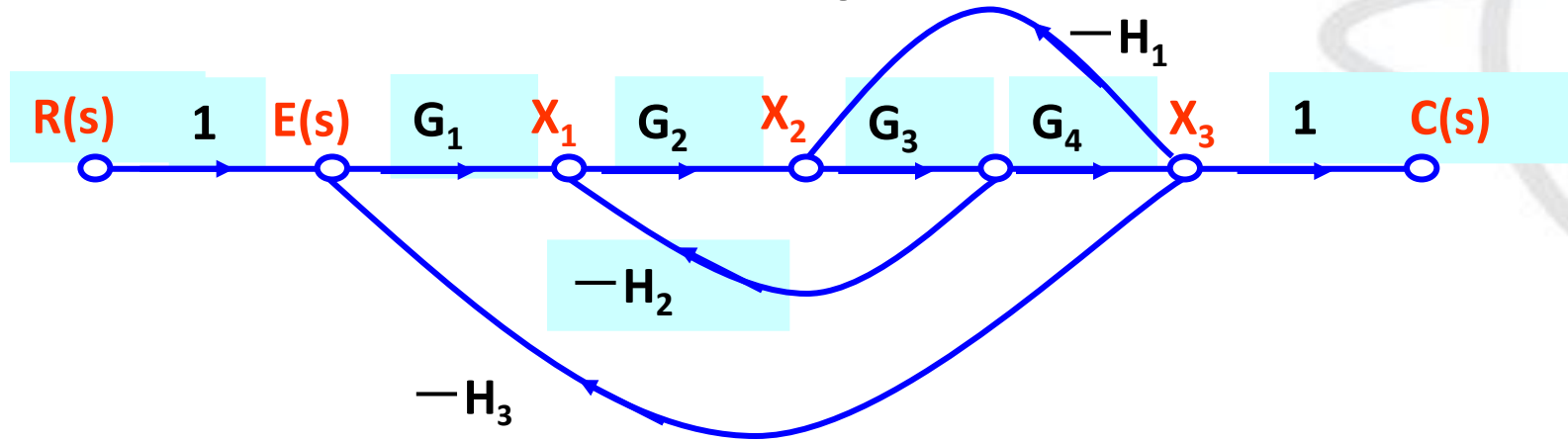
From Block Diagram to Signal-Flow Graph Models

Example#5



From Block Diagram to Signal-Flow Graph Models

Example#5

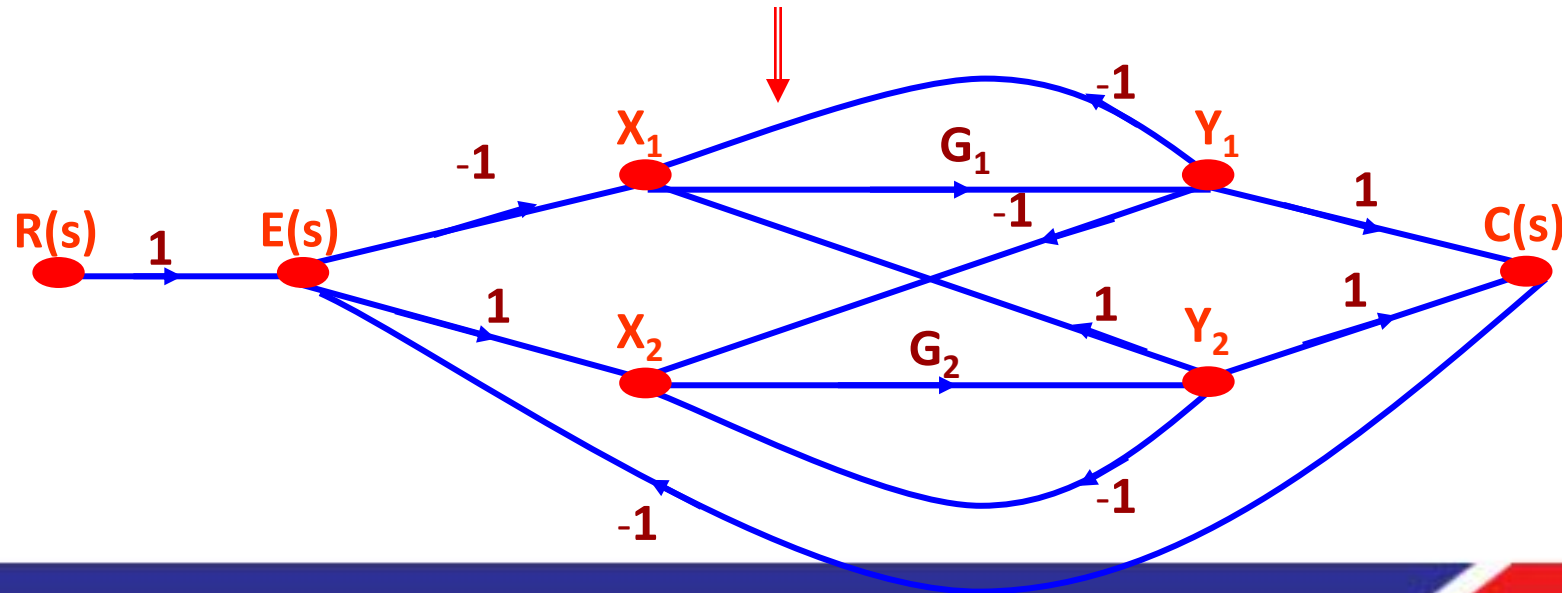
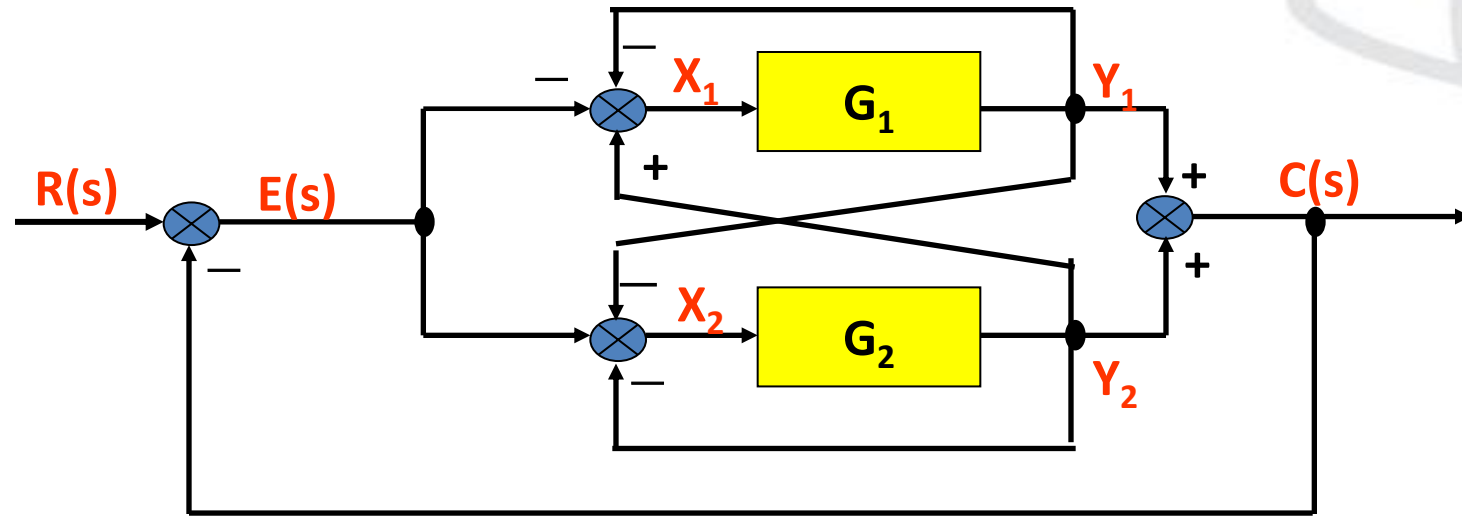


$$\Delta = 1 + (G_1 G_2 G_3 G_4 H_3 + G_2 G_3 H_2 + G_3 G_4 H_1)$$

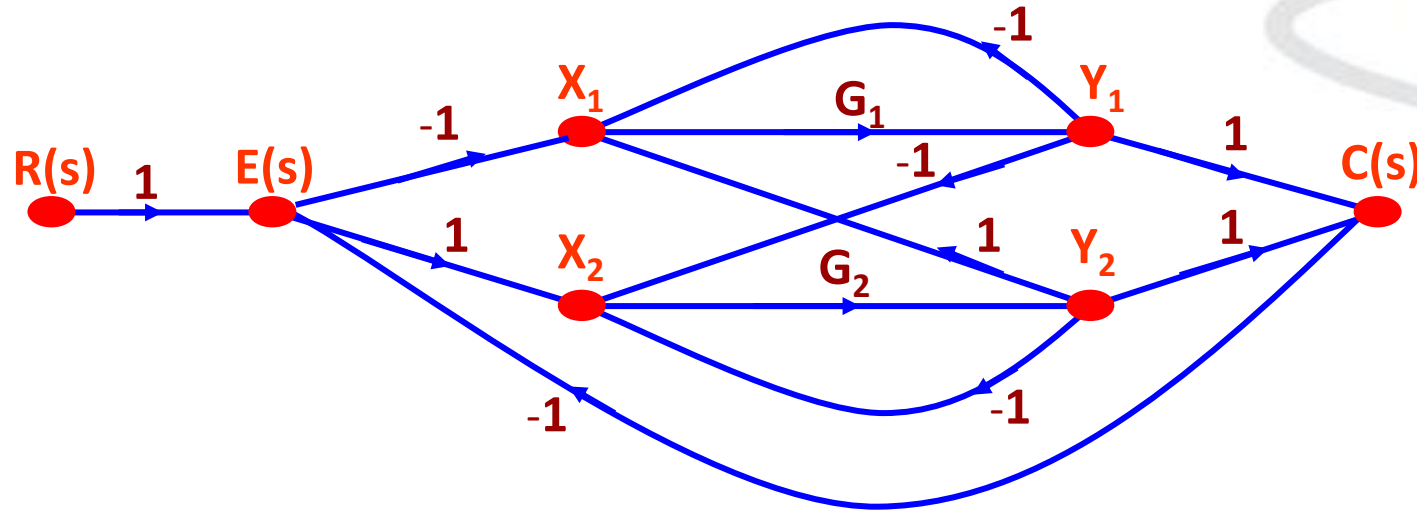
$$P_1 = G_1 G_2 G_3 G_4; \quad \Delta_1 = 1$$

$$G = \frac{C(s)}{R(s)} = \frac{G_1 G_2 G_3 G_4}{1 + G_1 G_2 G_3 G_4 H_3 + G_2 G_3 H_2 + G_3 G_4 H_1}$$

Example#6



Example#6



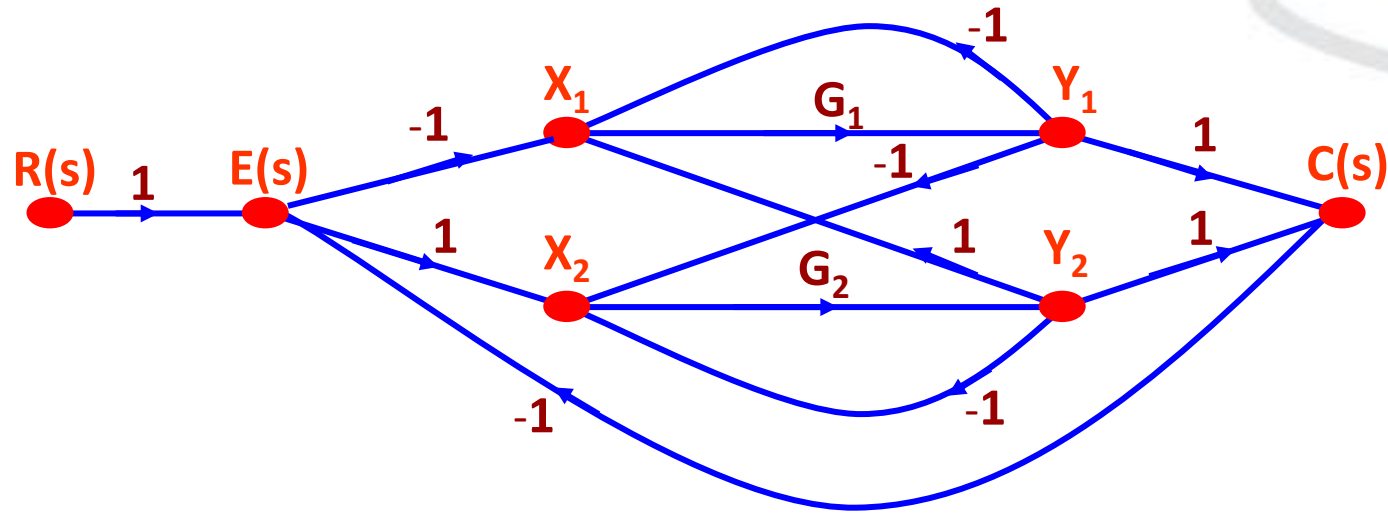
7 loops:

$$\begin{aligned}
 &[G_1 \cdot (-1)]; \quad [G_2 \cdot (-1)]; \quad [G_1 \cdot (-1) \cdot G_2 \cdot 1]; \quad [(-1) \cdot G_1 \cdot 1 \cdot (-1)]; \\
 &[(-1) \cdot G_1 \cdot (-1) \cdot G_2 \cdot 1 \cdot (-1)]; \quad [1 \cdot G_2 \cdot 1 \cdot (-1)]; \quad [1 \cdot G_2 \cdot 1 \cdot G_1 \cdot 1 \cdot (-1)].
 \end{aligned}$$

3 '2 non-touching loops' :

$$\begin{aligned}
 &[G_1 \cdot (-1)] \cdot [G_2 \cdot (-1)]; \quad [(-1) \cdot G_1 \cdot 1 \cdot (-1)] \cdot [G_2 \cdot (-1)]; \\
 &[1 \cdot G_2 \cdot 1 \cdot (-1)] \cdot [G_1 \cdot (-1)].
 \end{aligned}$$

Example#6



Then: $\Delta = 1 + 2G_2 + 4G_1G_2$

4 forward paths:

$$p_1 = (-1) \cdot G_1 \cdot 1 \quad \Delta_1 = 1 + G_2$$

$$p_2 = (-1) \cdot G_1 \cdot (-1) \cdot G_2 \cdot 1 \quad \Delta_2 = 1$$

$$p_3 = 1 \cdot G_2 \cdot 1 \quad \Delta_3 = 1 + G_1$$

$$p_4 = 1 \cdot G_2 \cdot 1 \cdot G_1 \cdot 1 \quad \Delta_4 = 1$$

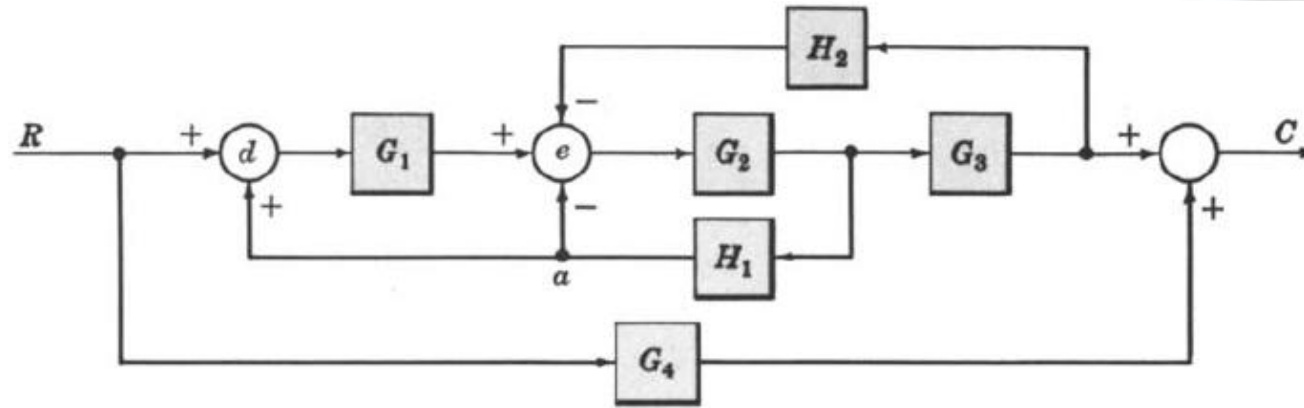
Example#6

We have

$$\begin{aligned}\frac{C(s)}{R(s)} &= \frac{\sum p_k \Delta_k}{\Delta} \\ &= \frac{G_2 - G_1 + 2G_1G_2}{1 + 2G_2 + 4G_1G_2}\end{aligned}$$



Example-7: Determine the transfer function C/R for the block diagram below by signal flow graph techniques.



- The signal flow graph of the above block diagram is shown below.

- There are two forward paths. The path gains are

$$P_1 = G_1 G_2 G_3 \text{ and } P_2 = G_4$$

- The three feedback loop gains are

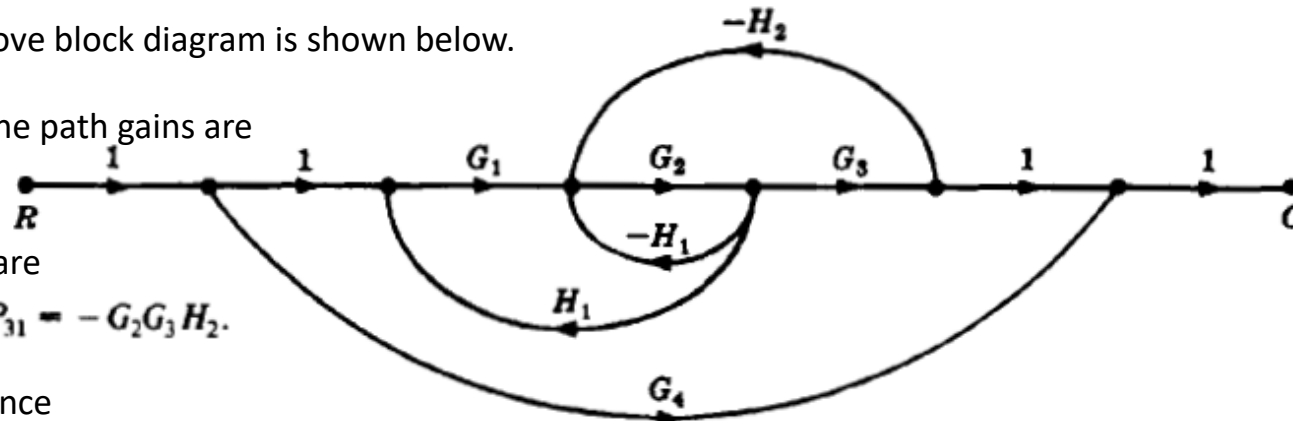
$$P_{11} = -G_2 H_1, P_{21} = G_1 G_2 H_1, P_{31} = -G_2 G_3 H_2.$$

- No loops are non-touching, hence

$$\Delta = 1 - (P_{11} + P_{21} + P_{31})$$

- Because the loops touch the nodes of P_1 , hence $\Delta_1 = 1$

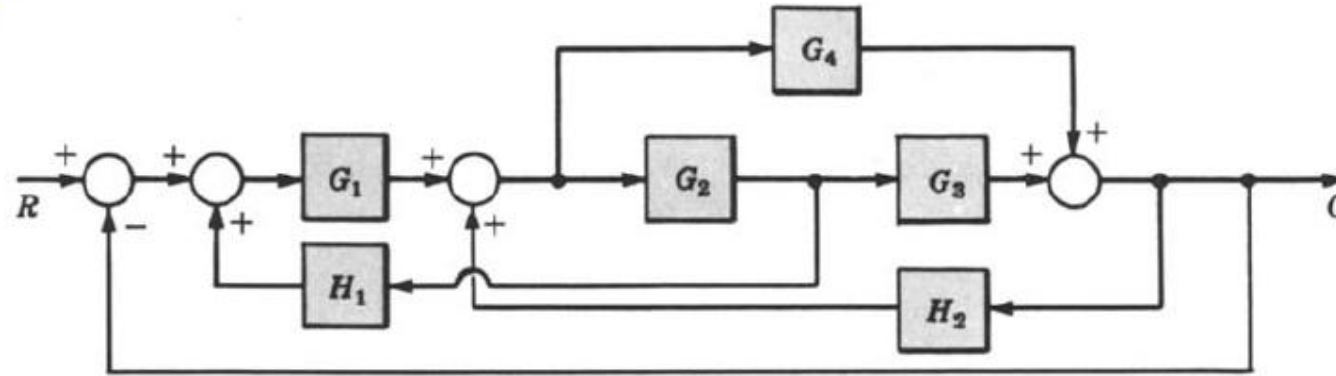
- Since no loops touch the nodes of P_2 , therefore $\Delta_2 = \Delta$.



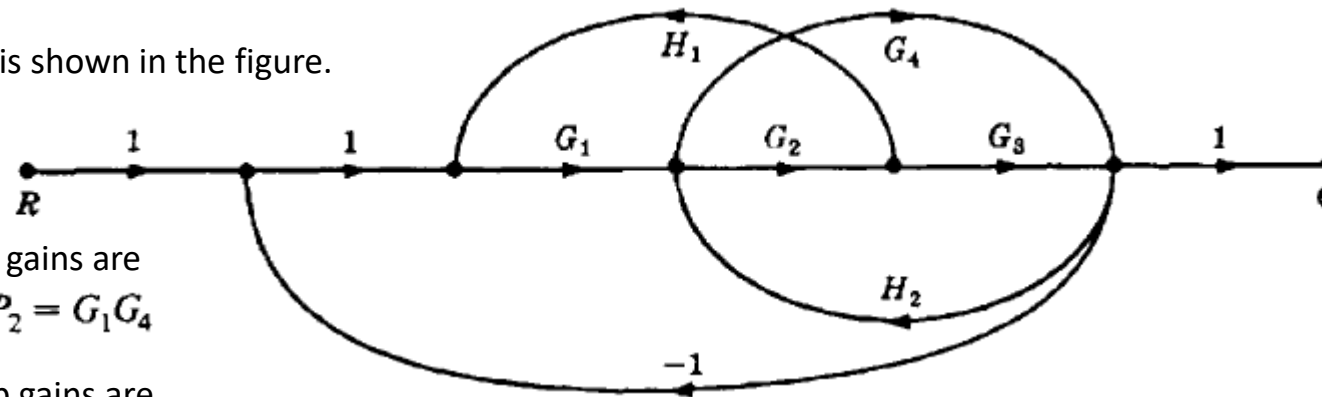
- Hence the control ratio $T = C/R$ is

$$T = \frac{P_1 \Delta_1 + P_2 \Delta_2}{\Delta} = \frac{G_1 G_2 G_3 + G_4 + G_2 G_4 H_1 - G_1 G_2 G_4 H_1 + G_2 G_3 G_4 H_2}{1 + G_2 H_1 - G_1 G_2 H_1 + G_2 G_3 H_2}$$

Example-6: Find the control ratio C/R for the system given below.



- The signal flow graph is shown in the figure.



- The two forward path gains are $P_1 = G_1 G_2 G_3$ and $P_2 = G_1 G_4$
- The five feedback loop gains are $P_{11} = G_1 G_2 H_1$, $P_{21} = G_2 G_3 H_2$, $P_{31} = -G_1 G_2 G_3$, $P_{41} = G_4 H_2$, and $P_{51} = -G_1 G_4$.
- All feedback loops touches the two forward paths, hence $\Delta_1 = \Delta_2 = 1$

- There are no non-touching loops, hence

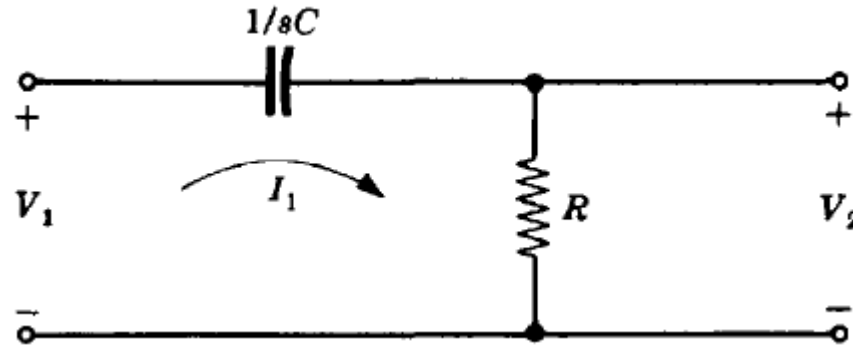
$$\Delta = 1 - (P_{11} + P_{21} + P_{31} + P_{41} + P_{51})$$

$$= 1 + G_1 G_2 G_3 - G_1 G_2 H_1 - G_2 G_3 H_2 - G_4 H_2 + G_1 G_4$$

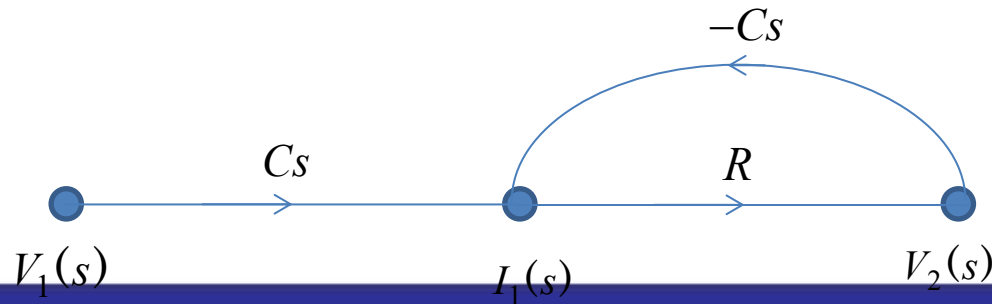
- Hence the control ratio $T =$

$$\frac{C}{R} = \frac{P_1 \Delta_1 + P_2 \Delta_2}{\Delta} = \frac{G_1 G_2 G_3 + G_1 G_4}{1 + G_1 G_2 G_3 - G_1 G_2 H_1 - G_2 G_3 H_2 - G_4 H_2 + G_1 G_4}$$

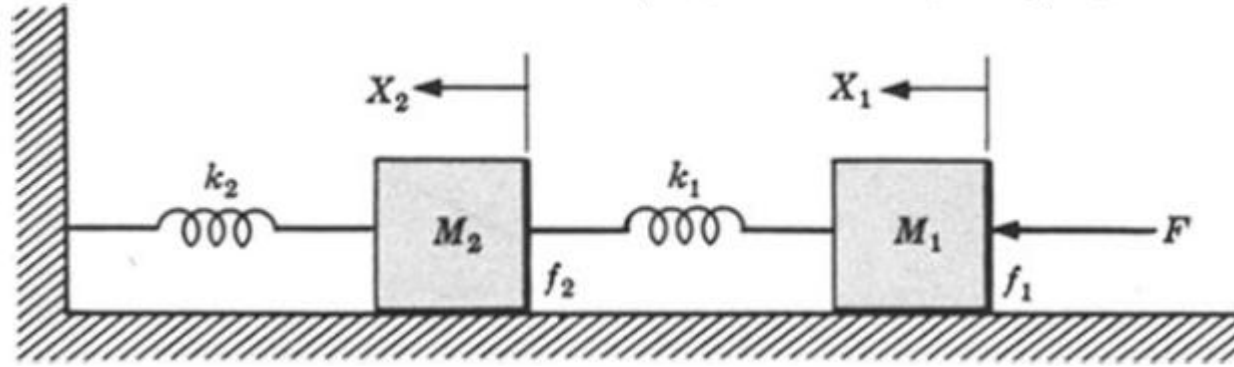
Design Example#1



$$\left. \begin{aligned} V_1(s) &= \frac{1}{Cs} I_1(s) + I_1(s)R \\ V_2(s) &= I_1(s)R \end{aligned} \right\} \Rightarrow CsV_1(s) - CsV_2(s) = I_1(s)$$



Design Example#2



$$F = M_1 s^2 X_1 + k_1 (X_1 - X_2) \quad 0 = M_2 s^2 X_2 + k_1 (X_2 - X_1) + k_2 X_2$$

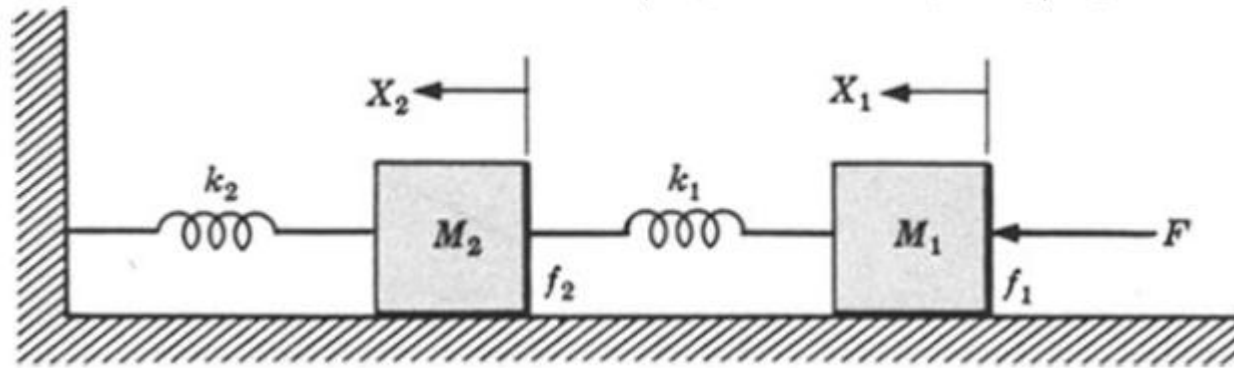
$$\begin{aligned} \text{(i)} \quad & F + k_1 X_2 = (M_1 s^2 + f_1 s + k_1) X_1 \\ \text{(ii)} \quad & k_1 X_1 = (M_2 s^2 + f_2 s + k_1 + k_2) X_2 \end{aligned}$$

There are three variables: X_1 , X_2 , and F . Therefore we need three nodes.

$$\text{(iii)} \quad \left(\frac{1}{A} \right) F + \left(\frac{k_1}{A} \right) X_2 = X_1$$

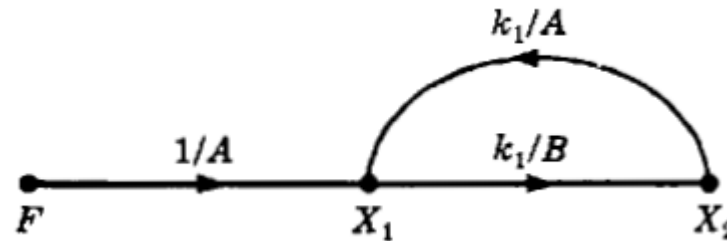
$$\text{(iv)} \quad \left(\frac{k_1}{B} \right) X_1 = X_2$$

Design Example#2

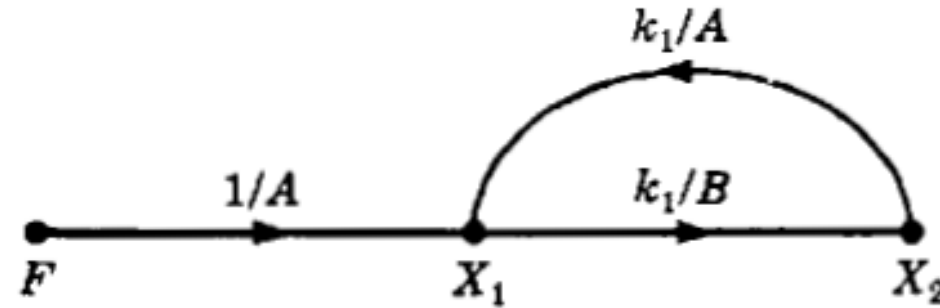


$$(iii) \quad \left(\frac{1}{A} \right) F + \left(\frac{k_1}{A} \right) X_2 = X_1$$

$$(iv) \quad \left(\frac{k_1}{B} \right) X_1 = X_2$$



Design Example#2



The forward path gain is $P_1 = k_1/AB$. The feedback loop gain is $P_{11} = k_1^2/AB$. then $\Delta = 1 - P_{11} = (AB - k_1^2)/AB$ and $\Delta_1 = 1$. Finally,

$$\frac{X_2}{F} = \frac{P_1 \Delta_1}{\Delta} = \frac{k_1}{AB - k_1^2} = \frac{k_1}{(M_1 s^2 + f_1 s + k_1)(M_2 s^2 + f_2 s + k_1 + k_2) - k_1^2}$$